

SortingLine_Project

Project

Name:	SortingLine_Project	Creation time:	09/28/2026 22:37:00
Last change	10/03/2026 21:41:06	Author:	HoraEmbedded
Last modified by:	ENVY	Version:	
Comment:			

Operating system

Name	Description
Operating system	Microsoft Windows 11 Pro
Version of the operating system	10.0.26300.0
Operating system service pack	
Version of the Internet Explorer	11.1882.26100.0
Computer name	DESKTOP-2HORA
User name	DESKTOP-2HORA\ENVY
Installation path of the TIA Portal	C:\Program Files\Siemens\Automation\Portal V18

Components

Name	Version	Release
AWB V2 - Automation License Manager Plugin V2.0 (AWB_V2)	V2.0	V02.00.00.00_01.04.00.03
AWB V2 - Software Management Plugin V2.0 (AWB_V2)	V2.0	V02.00.00.00_01.04.00.03
AWB V2 - TIA Addin V2.0 (AWB_V2)	V2.0	V02.00.00.00_01.04.00.03
AWB V2 - Central User Management Plugin V2.0 (AWB_V2)	V2.0	V02.00.00.00_01.04.00.03
AWB V2 - Host V2.1 (AWB_V2)	V2.1	02.01.00.00_01.00.00.50
S7-PLCSIM - S7-PLCSIM Setup V18.0 SP2 (PLCSIM_V18)	V18.0 + SP2	V18.00.02.00_09.13.00.01
TIA Portal Project Server - TIA Portal Project Server Single SetupPackage V1.1 (ProjectServer)	V1.1	V01.01.00.00_00.00.01.52
TIA Administrator - TIA Administrator V1.0 SP4 (TIAADMIN)	V1.0 + SP4	V01.00.04.00_01.18.00.04
Totally Integrated Automation Portal V18 - TIA Portal Single SetupPackage V18.0 SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - HM All Editions Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - HM NoBasic Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - Hardware Support Base Package 0 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Multiuser Client Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - Version Control Interface SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - STEP 7 Safety Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - STEP 7 Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01

Totally Integrated Automation Portal		
Name	Version	Release
Siemens Totally Integrated Automation Portal V18 - Hardware Support Base Package 02 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Hardware Support Base Package 03 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Hardware Support Base Package 04 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Support Base Package TO-01 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Support Base Package TO-02 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - Hardware Support Base Package WCF-01 V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_35.01.00.01
Siemens Totally Integrated Automation Portal V18 - TIACOMPCHCK Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - TIA Portal Security Audit Log Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - TIA Portal Shim Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - Simatic Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - WinCC Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - Openness SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - WinCC Transfer Mandatory Single SetupPackage V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
User Management Component - UserManagementComponentx64 V2.10 (UMC64)	V2.10	V02.10.00.00_00.00.00.38
User Management Component - umtrayiconx64 V2.10 (UMC64)	V2.10	V02.10.00.00_00.00.00.38
WinCC Runtime Advanced V17.0 - SIMATIC WinCC Runtime Advanced V17.0 (HMIRTM_V11)	V17.0 UPD5	V17.00.00.05_04.01.00.01
WinCC Runtime Advanced V17.0 - HMIRTM Tagging Package 01 Single SetupPackage V17.0 UPD5 (HMIRTM_V11)	V17.0 UPD5	V17.00.00.05_04.01.00.01
PLCSIM Advanced Single SetupPackage - PLCSIM Advanced Single SetupPackage V5.0 Upd2 (PLCSIMADV)	V5.0 + Upd2	V05.00.00.02_01.00.00.03
SIMATIC S7-PCT - SIMATIC S7-PCT V3.5 SP3 (S7PCT)	V3.5 + SP3	V03.05.03.00_01.12.00.13
WinCC Runtime Professional V18 - SIMATIC WinCC Runtime V18.0 (SCADA-RT_V11)	V18.0	V07.06.18.00_01.38.00.01
WinCC Runtime Professional V18 - OPCUA_Client V1.1 + SP3 (SCADA-RT_V11)	V1.1 + SP3	V01.01.03.00_01.09.00.03
WinCC Runtime Professional V18 - SCADA Simulation Single SetupPackage V18.0 + SP1 (SCADA-RT_V11)	V18.0 + SP1	V18.00.01.00_42.01.00.01
Siemens Totally Integrated Automation Portal V18 - Simatic Single SetupPackage 32 Bit V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01

Totally Integrated Automation Portal		
Name	Version	Release
Siemens Totally Integrated Automation Portal V18 - WinCC Single SetupPackage 32 Bit V18.0 + SP1 (TIAP18)	V18.0 + SP1	V18.00.01.00_42.01.00.01
AddinRolloutService	18.0.1.0	V18.00.01.00_42.01.00.01
SIMATIC HMI License Manager Panel Plugin (x64)	18.0.1.0	V18.00.01.00_42.01.00.01
Automation Access Control Component x64	05.01	K05.01.02.00_00.00.00.02
SIMATIC WinCC Runtime Advanced Driver (x64)	18.0.1.0	V18.00.01.00_42.01.00.01
ETWEventCollector	18.0.1.0	V18.00.01.00_42.01.00.01
Installation Integrity Validator	1.12.18.0	V01.12.18.00_00.00.00.00
SIMATIC NCM FWL 64	5.6.0.3	K5.6.0.3_1.1.0.2
NCM GPRS 64	01.02.00.00	V1.2.0.0_2.1.0.1
SIMATIC PLCSIM 64	17.00.00	17.00.00.00_01.00.37.02
SIMATIC PLCSIM Advanced Driver64	5.0.0.2	V05.00.00.02_01.00.00.03
SIMATIC Device Drivers	9.3	09.03.04.00_01.01.00.37
TelemetryConnector	1.8.0.49	V01.08.00.49_01.00.00.00
Automation Access Control Component	05.01	K05.01.02.00_00.00.00.02
SIMATIC HMIProvider	7.0	K07.00.03.01_01.01.00.01
SIEMENS OPC	3.9	03.09.12.00_01.07.00.01
SIMATIC PLCSIM Advanced SimRT	5.0.0.2	V05.00.00.02_01.00.00.03
SIMATIC HMI ProSave	18.0.1.0	V18.00.01.00_42.01.00.01
SIMATIC HMI Symbol Library	17.0.0.5	V17.00.00.05_04.01.00.01
SIMATIC HMI Touch Input	17.0.0.5	V17.00.00.05_04.01.00.01
SIMATIC Runtime Interfaces	2.1	K02.01.00.03_01.01.00.01
SIMATIC Device Drivers WoW	29.3	29.03.04.00_01.01.00.37
SIMATIC Event Database	5.6	05.06.02.03_02.01.00.01
SeCon	2.9	V02.09.00.00_01.03.00.01
SIMATIC Station Observer	K7.3.1.0	V07.03.01.00_01.01.00.14
SIMATIC SCS	V8.0.0.0	V08.00.00.00_01.09.00.10
SIMATIC WinCC Common Archiving	V8.0.0.0	V07.06.18.00_01.38.00.01
WinCC Runtime Advanced Simulator	17.0.0.0	V17.00.00.00_43.02.00.01
Products		
Name	Version	Release
TIA Administrator	V2.0	V02.00.00.00_01.04.00.03
S7-PLCSIM	V18 SP2	18.00.02.00_09.13.00.01
TIA Portal Project Server	V1.1	V01.01.00.00_00.00.01.52
SIMATIC STEP 7 Prof - STEP 7 Safety - WinCC Adv	V17.0	V17.00.00.00_43.02.00.01
SIMATIC STEP 7 Prof - STEP 7 Safety - WinCC Prof	V18.0	V18.00.01.00_42.01.00.01
User Management Component	V2.10	V02.10.00.00_00.00.00.38
UMC Status Application	V2.10	V02.10.00.00_00.00.00.38
SIMATIC WinCC Runtime Advanced Simulation	V17.0 Upd5	V17.00.00.05_04.01.00.01
S7-PLCSIM Advanced	V5.0 Upd2	V05.00.00.02_01.00.00.03
S7-PCT	V3.5 SP3	V03.05.03.00_01.12.00.13
SIMATIC WinCC Runtime Professional Simulation	V18.0	V18.00.01.00_42.01.00.01
Automation License Manager	V6.2 + SP5	06.02.05.00_00.00.03.23
S7-PLCSIM	V5.4 + SP8 + Upd2	V05.04.08.02_02.40.00.01
SIMATIC ProSave	V18.0	V18.00.01.00_42.01.00.01
S7-PCT	V3.5 SP3	K3.5.3.0_1.12.0.13

Totally Integrated Automation Portal			
SortingLine_Project PLC_Sorting [CPU 1214C DC/DC/DC]			
PLC_Sorting			
General\Project information			
Name	PLC_Sorting	Author	HoraEmbedded
Comment		Slot	1
Rack	0		
General\Catalog information			
Short designation	CPU 1214C DC/DC/DC	Description	Work memory 150 KB; 24VDC power supply with DI14 x 24VDC SINK/ SOURCE, DQ10 x 24VDC and AI2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA
Article number	6ES7 214-1AG40-0XB0	Firmware version	V4.6
	False		
General\Identification & Maintenance			
Plant designation		Location identifier	
Installation date	2026-09-28 22:38:48.935	Additional information	
General\Checksums			
Text lists	FA 70 E8 75 1D 5A 8E 29	Software	Not available (compile necessary)
PROFINET interface [X1]\General			
Name	PROFINET interface_1	Author	HoraEmbedded
Comment			
PROFINET interface [X1]\General\Project information			
Name	DI 14/DQ 10_1	Comment	
Name	AI 2_1	Comment	
PROFINET interface [X1]\Ethernet addresses\Interface networked with			
Subnet:	Not connected		
PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)			
IP configuration	Set IP address in the project	IP address:	192.168.0.1
Subnet mask:	255.255.255.0	Use router	False
PROFINET interface [X1]\Ethernet addresses\PROFINET			
PROFINET device name is set directly at the device	False	Generate PROFINET device name automatically	True
PROFINET device name:	plc_sorting	Converted name:	plcxbsorting9923
Device number:	0		
PROFINET interface [X1]\Time synchronization			
Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses
Server 1	0.0.0.0	Server 2	0.0.0.0
Server 3	0.0.0.0	Server 4	0.0.0.0
Update interval	10sec		
CPU synchronizes the modules of the device.	No synchronization		

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PROFINET interface [X1]\Digital inputs\Channel0					
Channel address	I100.0	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel0\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49152		
Event name:	0	Hardware interrupt:	0		
Rising edge0	Rising edge0				
PROFINET interface [X1]\Digital inputs\Channel0\					
Enable falling edge detection	0	Prefix Event Falling Edge	49280		
Event name:	0	Hardware interrupt:	0		
Falling edge0	Falling edge0				
PROFINET interface [X1]\Digital inputs\Channel1					
Channel address	I100.1	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel1\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49153		
Event name:	0	Hardware interrupt:	0		
Rising edge1	Rising edge1				
PROFINET interface [X1]\Digital inputs\Channel1\					
Enable falling edge detection	0	Prefix Event Falling Edge	49281		
Event name:	0	Hardware interrupt:	0		
Falling edge1	Falling edge1				
PROFINET interface [X1]\Digital inputs\Channel2					
Channel address	I100.2	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel2\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49154		
Event name:	0	Hardware interrupt:	0		
Rising edge2	Rising edge2				
PROFINET interface [X1]\Digital inputs\Channel2\					
Enable falling edge detection	0	Prefix Event Falling Edge	49282		
Event name:	0	Hardware interrupt:	0		
Falling edge2	Falling edge2				
PROFINET interface [X1]\Digital inputs\Channel3					
Channel address	I100.3	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel3\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49155		
Event name:	0	Hardware interrupt:	0		
Rising edge3	Rising edge3				
PROFINET interface [X1]\Digital inputs\Channel3\					
Enable falling edge detection	0	Prefix Event Falling Edge	49283		
Event name:	0	Hardware interrupt:	0		
Falling edge3	Falling edge3				
PROFINET interface [X1]\Digital inputs\Channel4					
Channel address	I100.4	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel4\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49156		
Event name:	0	Hardware interrupt:	0		

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Rising edge4	Rising edge4		
PROFINET interface [X1]\Digital inputs\Channel4\			
Enable falling edge detection	0	Prefix Event Falling Edge	49284
Event name:	0	Hardware interrupt:	0
Falling edge4	Falling edge4		
PROFINET interface [X1]\Digital inputs\Channel5			
Channel address	I100.5	Input filters	6.4 millise
Enable pulse catch	0		
PROFINET interface [X1]\Digital inputs\Channel5\			
Enable rising edge detection	0	Prefix Event Rising Edge	49157
Event name:	0	Hardware interrupt:	0
Rising edge5	Rising edge5		
PROFINET interface [X1]\Digital inputs\Channel5\			
Enable falling edge detection	0	Prefix Event Falling Edge	49285
Event name:	0	Hardware interrupt:	0
Falling edge5	Falling edge5		
PROFINET interface [X1]\Digital inputs\Channel6			
Channel address	I100.6	Input filters	6.4 millise
Enable pulse catch	0		
PROFINET interface [X1]\Digital inputs\Channel6\			
Enable rising edge detection	0	Prefix Event Rising Edge	49158
Event name:	0	Hardware interrupt:	0
Rising edge6	Rising edge6		
PROFINET interface [X1]\Digital inputs\Channel6\			
Enable falling edge detection	0	Prefix Event Falling Edge	49286
Event name:	0	Hardware interrupt:	0
Falling edge6	Falling edge6		
PROFINET interface [X1]\Digital inputs\Channel7			
Channel address	I100.7	Input filters	6.4 millise
Enable pulse catch	0		
PROFINET interface [X1]\Digital inputs\Channel7\			
Enable rising edge detection	0	Prefix Event Rising Edge	49159
Event name:	0	Hardware interrupt:	0
Rising edge7	Rising edge7		
PROFINET interface [X1]\Digital inputs\Channel7\			
Enable falling edge detection	0	Prefix Event Falling Edge	49287
Event name:	0	Hardware interrupt:	0
Falling edge7	Falling edge7		
PROFINET interface [X1]\Digital inputs\Channel8			
Channel address	I101.0	Input filters	6.4 millise
Enable pulse catch	0		
PROFINET interface [X1]\Digital inputs\Channel8\			
Enable rising edge detection	0	Prefix Event Rising Edge	49160
Event name:	0	Hardware interrupt:	0
Rising edge8	Rising edge8		
PROFINET interface [X1]\Digital inputs\Channel8\			
Enable falling edge detection	0	Prefix Event Falling Edge	49288
Event name:	0	Hardware interrupt:	0
Falling edge8	Falling edge8		

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PROFINET interface [X1]\Digital inputs\Channel9					
Channel address	I101.1	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel9\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49161		
Event name:	0	Hardware interrupt:	0		
Rising edge9	Rising edge9				
PROFINET interface [X1]\Digital inputs\Channel9\					
Enable falling edge detection	0	Prefix Event Falling Edge	49289		
Event name:	0	Hardware interrupt:	0		
Falling edge9	Falling edge9				
PROFINET interface [X1]\Digital inputs\Channel10					
Channel address	I101.2	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel10\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49162		
Event name:	0	Hardware interrupt:	0		
Rising edge10	Rising edge10				
PROFINET interface [X1]\Digital inputs\Channel10\					
Enable falling edge detection	0	Prefix Event Falling Edge	49290		
Event name:	0	Hardware interrupt:	0		
Falling edge10	Falling edge10				
PROFINET interface [X1]\Digital inputs\Channel11					
Channel address	I101.3	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel11\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49163		
Event name:	0	Hardware interrupt:	0		
Rising edge11	Rising edge11				
PROFINET interface [X1]\Digital inputs\Channel11\					
Enable falling edge detection	0	Prefix Event Falling Edge	49291		
Event name:	0	Hardware interrupt:	0		
Falling edge11	Falling edge11				
PROFINET interface [X1]\Digital inputs\Channel12					
Channel address	I101.4	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel13					
Channel address	I101.5	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	Measurement type	Voltage		
Voltage range	0..10 V	Smoothing	Weak (4 cycles)		
		Enable overflow diag- nostics	1		
PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	Measurement type	Voltage		
Voltage range	0..10 V	Smoothing	Weak (4 cycles)		
		Enable overflow diag- nostics	1		
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				

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PROFINET interface [X1]\Digital outputs\Channel0			
Channel address	Q0.0	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel1			
Channel address	Q0.1	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel2			
Channel address	Q0.2	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel3			
Channel address	Q0.3	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel4			
Channel address	Q0.4	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel5			
Channel address	Q0.5	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel6			
Channel address	Q0.6	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel7			
Channel address	Q0.7	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel8			
Channel address	Q1.0	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Digital outputs\Channel9			
Channel address	Q1.1	Substitute a value of 1 on a change from RUN to STOP.	0
PROFINET interface [X1]\Operating mode			
IO controller	True	IO system	
Device number	0	IO device	False
PROFINET interface [X1]\I/O addresses\Input addresses			
Start address	100.0	End address	101.7
Organization block	0	Process image	0
PROFINET interface [X1]\I/O addresses\Input addresses			
Start address	64	End address	67
Organization block	0	Process image	0
PROFINET interface [X1]\I/O addresses\Output addresses			
Start address	0.0	End address	1.7
Organization block	0	Process image	0
PROFINET interface [X1]\Advanced options\Interface options			
Support device replacement without exchangeable medium	True	Permit overwriting of device names of all assigned IO devices	False
Use IEC V2.2 LLDP mode	False	Keep-Alive connection monitoring:	30s
PROFINET interface [X1]\Advanced options\Real time settings\IO communication			
Send clock:	1.000ms		

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PROFINET interface [X1]\Advanced options\Real time settings\Real time options			
Calculated bandwidth for cyclic IO data:	0.000ms	Calculated bandwidth for cyclic IO data:	0.000%
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General			
Name	Port_1	Author	ENVY
Comment			
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:			
Local port:	PLC_Sorting\PROFINET interface_1 [X1]\Port_1 [X1 P1]	Medium:	Copper
Cable name:	---		
			
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:			
	Monitoring of partner port is not possible	Partner port:	Any partner
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate			
Activate this port for use	True		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection			
Transmission rate / duplex:	Automatic	Monitor	False
Enable autonegotiation	True		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries			
End of detection of accessible devices	False	End of topology discovery	False
End of the sync domain	False		
PROFINET interface [X1]\Web server access			
Enable Web server for the IP address of this interface	False	The Web server must also be activated in the properties of the PLC.	
High speed counters (HSC)\HSC1\General\Enable			
Enable this high speed counter	0	Enable this high speed counter	0
Enable this high speed counter	0	Enable this high speed counter	0
Enable this high speed counter	0	Enable this high speed counter	0
High speed counters (HSC)\HSC1\General\Project information			
Name	HSC_1	Comment	
Name	HSC_2	Comment	
Name	HSC_3	Comment	
Name	HSC_4	Comment	
Name	HSC_5	Comment	
Name	HSC_6	Comment	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses			
Start address	1000.0	End address	1003.7
Start address	1004.0	End address	1007.7
Organization block	0	Start address	1008.0
End address	1011.7	Organization block	0
Process image	0	Start address	1012.0

Totally Integrated Automation Portal					
End address	1015.7	Organization block	0		
Process image	0	Start address	1016.0		
End address	1019.7	Organization block	0		
Process image	0	Start address	1020.0		
End address	1023.7	Organization block	0		
Process image	0	Organization block	0		
Process image	0	Process image	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable					
Enable this pulse generator	0	Enable this pulse generator	0		
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information					
Name	Pulse_1	Comment			
Name	Pulse_2	Comment			
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses					
Start address	1000.0	End address	1001.7		
Start address	1002.0	End address	1003.7		
Organization block	0	Organization block	0		
Process image	0	Process image	0		
Startup					
Startup after POWER ON	Warm restart - mode before POWER OFF	Comparison preset to actual configuration	Startup CPU even if mismatch		
Configuration time	60000ms	OBs should be interruptible	1		
Cycle					
Cycle monitoring time [ms]	150ms				
Enable minimum cycle time for cyclic OBs	0	Minimum cycle time	1ms		
Communication load					
Cycle load due to communication [%]	20%				
System and clock memory\System memory bits					
Enable the use of system memory byte	0	Address of system memory byte (MBx)	1		
First cycle		Diagnostic status changed			
Always 1 (high)		Always 0 (low)			
System and clock memory\Clock memory bits					
Enable the use of clock memory byte	0	Address of clock memory byte (MBx)	0		
10 Hz clock		5 Hz clock			
2.5 Hz clock		2 Hz clock			
1.25 Hz clock		1 Hz clock			
0.625 Hz clock		0.5 Hz clock			
Web server\General					
Activate Web server on all modules of this device	False	Permit access only with HTTPS	True		
Web server\Automatic update					
Enable automatic update	True	Update interval	0s		
Web server\User management					
User name		User rights			
Everybody					
Web server\User-defined web pages					
Application name	HTML source path	Default HTML page	Files with dynamic content	Web DB number	Fragment DB number
		index.htm	.htm;.html	333	334

Totally Integrated Automation Portal			
Web server\Overview of interfaces			
Device		Interface	Enabled web server access
PLC_Sorting		PROFINET interface_1	False
User interface languages			
Assign project language		User interface languages	
English (United States)		German	
English (United States)		English	
English (United States)		French	
English (United States)		Spanish	
English (United States)		Italian	
English (United States)		Chinese (simplified)	
Time of day\Local time			
Time zone		(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna	
Time of day\Daylight saving time			
Activate daylight saving time		1	Difference between standard and daylight saving time 60min
Time of day\Daylight saving time\Start of daylight saving time			
Starting week of the month:		Last	Sunday
in		March	at 1:00 a.m.
Time of day\Daylight saving time\Start of standard time			
		Last	Sunday
in		October	at 2:00 a.m.
Protection & Security			
Level of protection		No protection	
Protection & Security\Connection mechanisms			
Permit access with PUT/GET communication from remote partner		True	
Protection & Security\Security event			
Summarize diagnostics in case of high message volume		True	Length of an interval 20
Unit		seconds	
Protection & Security\External load memory			
Disable copying from internal load memory to external load memory		False	
Advanced configuration\DnsParameterConfigurationMenu			
The Tree-Node DnsConfigurationMenu was not filled by some ACF			
Advanced configuration\Configuration control\Configuration control for central configuration			
Allow to reconfigure the device via the user program		0	
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).			
Activate SNMP		False	
Configuration control\Configuration control for central configuration			
Allow to reconfigure the device via the user program		0	

Totally Integrated Automation Portal				
Connection resources\				
	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_Sorting [CPU 1214C DC/DC/DC] - Configured
Maximum number of resources:		34	34	68
	Maximum	Configured	Configured	Configured
PG communication:	4	-	-	-
HMI communication:	12	0	0	0
S7 communication:	8	0	0	0
Open user communication:	8	0	0	0
Web communication:	2	-	-	-
OPC UA client/server communication:	0	-	-	-
Other communication:	-	-	0	0
Total resources used:		0	0	0
Available resources:		34	34	68
Overview of addresses\Overview of addresses\Overview of addresses				
Inputs	True		Outputs	True
Address gaps	False		Slot	True

Totally Integrated Automation Portal										
Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	64	67	AI 2_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 35
I	100	101	DI 14/DQ 10_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 1

Totally Integrated Automation Portal

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Program blocks

Main [OB1]

Main Properties

General

Name	Main	Number	1	Type	OB
Language	LAD	Numbering	Automatic		

Information

Title	Main Program Cycle Execution	Author		Comment	Main cyclic block of the S7-1200 PLC. This block coordinates the execution of the entire digital twin program at every PLC scan cycle. It handles the mode manager block to evaluate safety states, runs the main automatic sorting logic sequence, and executes the centralized output gating block to control the physical actuators of the Factory I/O scene.
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

Network 1:

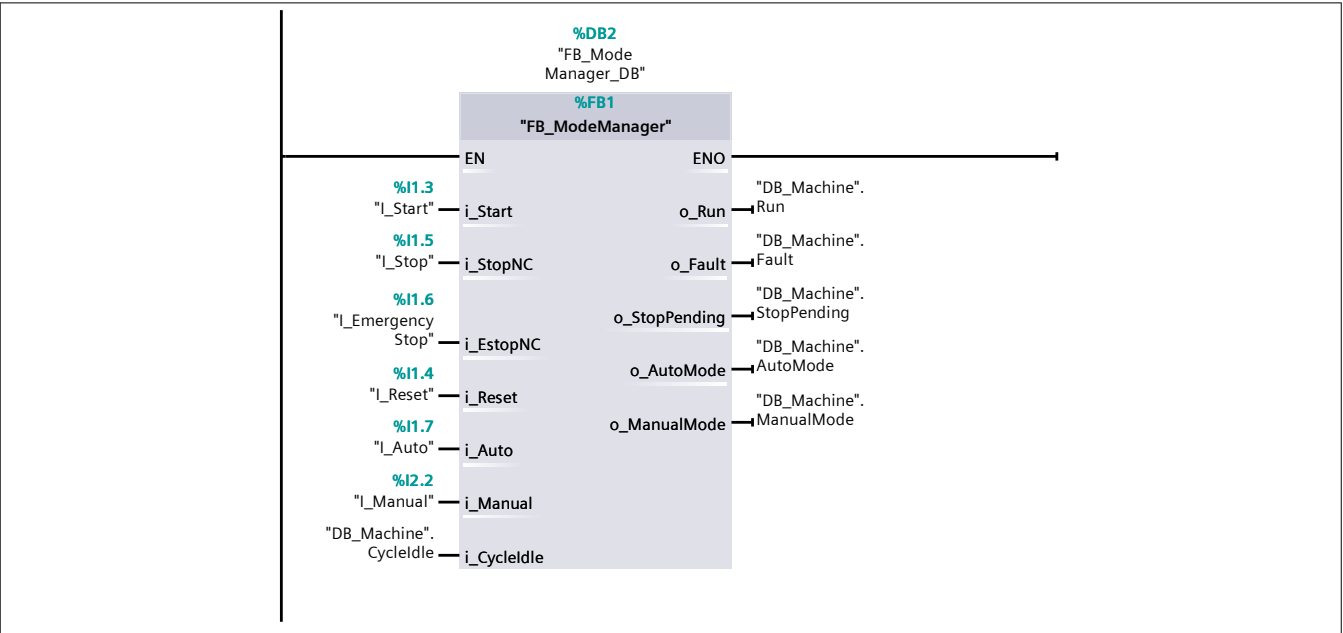
%FC9000

"MHJ-PLC-Lab-Function-S71200"

EN

ENO

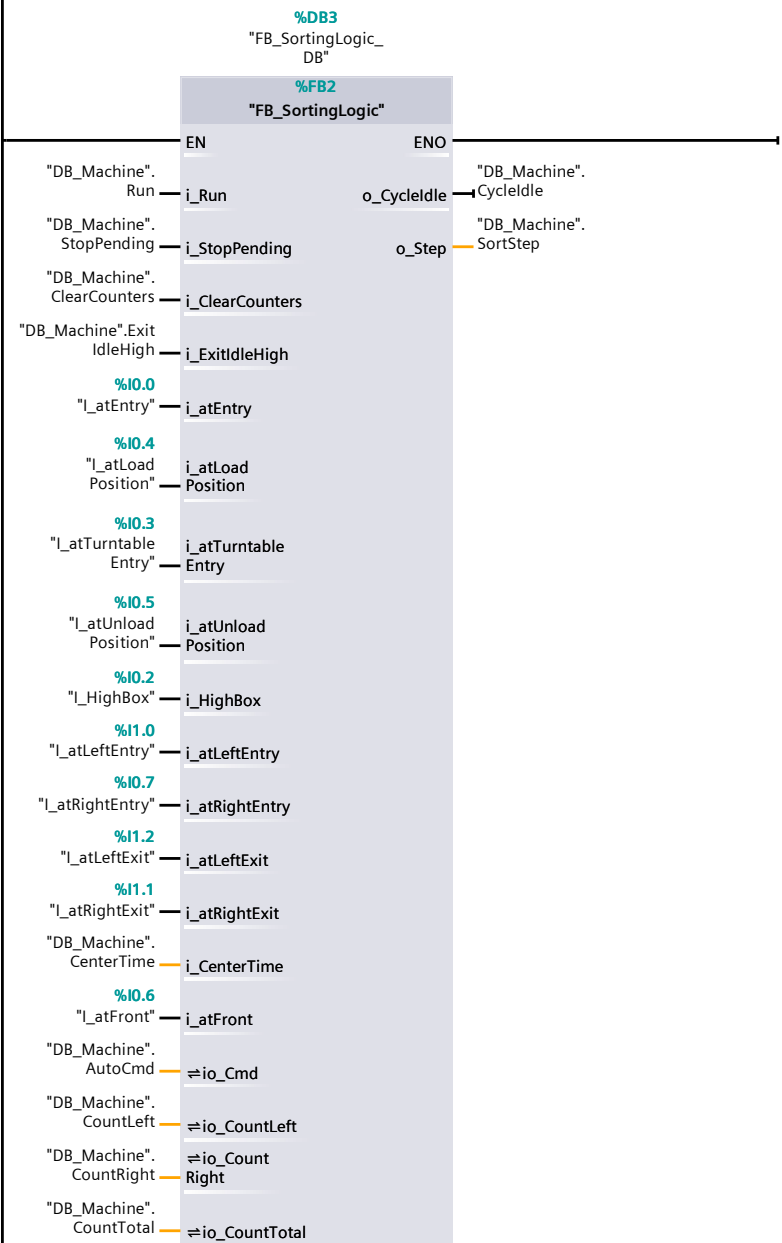
Network 2:



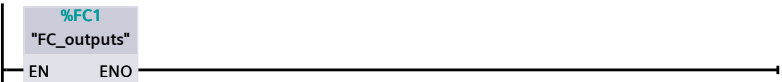
Network 3:



Network 4:



Network 5:



SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks

MHJ-PLC-Lab-Function-S71200 [FC9000]

MHJ-PLC-Lab-Function-S71200 Properties

General

Name	MHJ-PLC-Lab-Function-S71200	Number	9000	Type	FC
Language	SCL	Numbering	Manual		

Information

Title		Author	HoraEmbedded	Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Comment
Input			
Output			
InOut			
▼ Temp			
rdTimeReturn	Int		
▼ outputTime	DTL		
YEAR	UInt		
MONTH	USInt		
DAY	USInt		
WEEKDAY	USInt		
HOURL	USInt		
MINUTE	USInt		
SECOND	USInt		
NANOSECOND	UDInt		
SyncVal	Byte		
forVal	Int		
forVal_2	Int		
Value	Byte		
▼ Constant			
CompVal	Byte	16#34	
Value_01	Byte	16#11	
Value_01_DW	DWord	16#A165_D992	
Value_02_DW	DWord	16#58BE_4401	
▼ Return			
MHJ-PLC-Lab-Function-S71200	Void		

```
0001
0002 #Value:=PEEK(area := 16#82,
0003     dbNumber := 0,
0004     byteOffset := 511);
0005 #Value := #Value + 1;
0006
0007 POKE(area := 16#82,
0008     dbNumber := 0,
0009     byteOffset := 511,
0010     value := #Value);
0011
```

```
0012 POKE(area:=16#81,
0013     dbNumber:=0,
0014     byteOffset:=1016,
0015     value:=#Value_01_DW);
0016 POKE(area := 16#81,
0017     dbNumber := 0,
0018     byteOffset := 1020,
0019     value := #Value_02_DW);
0020
0021 POKE(area := 16#81,
0022     dbNumber := 0,
0023     byteOffset := 511,
0024     value := B#16#00);
0025
0026 FOR #forVal := 0 TO 120 DO
0027     FOR #forVal_2:=0 TO 10 DO
0028         #rdTimeReturn:=RD_SYS_T(#outputTime);
0029         #rdTimeReturn := WR_SYS_T(#outputTime);
0030         #rdTimeReturn := RD_SYS_T(#outputTime);
0031         #rdTimeReturn := WR_SYS_T(#outputTime);
0032     END_FOR;
0033     #SyncVal:= PEEK(area := 16#81,
0034         dbNumber := 0,
0035         byteOffset := 511);
0036     IF #SyncVal = #CompVal THEN
0037         GOTO M_1;
0038     END_IF;
0039 END_FOR;
0040 RETURN;
0041
0042 M_1:
0043 POKE(area := 16#81,
0044     dbNumber := 0,
0045     byteOffset := 511,
0046     value := B#16#0);
0047
0048
0049
```

Symbol	Address	Type	Comment
#CompVal	16#34	Byte	
#forVal		Int	
#forVal_2		Int	
#outputTime		DTL	
#rdTimeReturn		Int	
#SyncVal		Byte	
#Value		Byte	
#Value_01_DW	16#A165_D992	DWord	
#Value_02_DW	16#58BE_4401	DWord	

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks

DB_Machine [DB1]	
------------------	--

DB_Machine Properties

General	
1	General
2	General
3	General
4	General
5	General
6	General
7	General
8	General
9	General
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12	General
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93	General
94	General
95	General
96	General
97	General
98	General
99	General
100	General

Name	DB_Machine	Number	1	Type	DB
Language	DB	Numbering	Automatic		

Information

Title	Global Machine Status and Shared Data Registers	Author	HoraEmbedded	Comment	Global Data Block (DB) serving as the main internal memory exchange registry for the system. Contains structured variables for system operating flags (Run, Fault, Modes), active part counters (Total, Left, Right), manual control parameters generated by the operator panel, and live engineering settings such as the box centering delay time variable.
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain	Accessible from HMI/O PC UA/Web API	Writable from HMI/O PC UA/Web API	Visible in HMI engineering	Set-point	Supervision	Comment
▼ Static									
Run	Bool	false	False	True	True	True	False		
Fault	Bool	false	False	True	True	True	False		
StopPending	Bool	false	False	True	True	True	False		
AutoMode	Bool	false	False	True	True	True	False		
ManualMode	Bool	false	False	True	True	True	False		
CycleIdle	Bool	false	False	True	True	True	False		
CountLeft	DInt	0	False	True	True	True	False		
CountRight	DInt	0	False	True	True	True	False		
CountTotal	DInt	0	False	True	True	True	False		

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Totally Integrated Automation Portal										
Name	Data type	Start value	Retain	Access- ible from HMI/O PC UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion	Comment	
▼ AutoCmd	"UDT_ActuatorCmd"		False	True	True	True	False			
EntryConveyor	Bool	false	False	True	True	True	False			
FeederConveyor	Bool	false	False	True	True	True	False			
LeftConveyor	Bool	false	False	True	True	True	False			
RightConveyor	Bool	false	False	True	True	True	False			
Load	Bool	false	False	True	True	True	False			
Unload	Bool	false	False	True	True	True	False			
Turn	Bool	false	False	True	True	True	False			
Emit	Bool	false	False	True	True	True	False			
RemoverLeft	Bool	true	False	True	True	True	False			
RemoverRight	Bool	true	False	True	True	True	False			
▼ ManCmd	"UDT_ActuatorCmd"		False	True	True	True	False			
EntryConveyor	Bool	false	False	True	True	True	False			
FeederConveyor	Bool	false	False	True	True	True	False			
LeftConveyor	Bool	false	False	True	True	True	False			
RightConveyor	Bool	false	False	True	True	True	False			
Load	Bool	false	False	True	True	True	False			
Unload	Bool	false	False	True	True	True	False			
Turn	Bool	false	False	True	True	True	False			
Emit	Bool	false	False	True	True	True	False			
RemoverLeft	Bool	true	False	True	True	True	False			
RemoverRight	Bool	true	False	True	True	True	False			
SortStep	DInt	0	False	True	True	True	True			
ExitIdleHigh	Bool	false	False	True	True	True	False			

Totally Integrated Automation Portal											
Name	Data type	Start value	Retain	Access- sible from HMI/O PC UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion			Comment
ClearCounters	Bool	false	False	True	True	True	False				
CenterTime	Time	T#2s	False	True	True	True	False				

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Program blocks

FB_ModeManager [FB1]

FB_ModeManager Properties

General	
1	General
2	General
3	General
4	General
5	General
6	General
7	General
8	General
9	General
10	General
11	General
12	General
13	General
14	General
15	General
16	General
17	General
18	General
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83	General
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85	General
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89	General
90	General
91	General
92	General
93	General
94	General
95	General
96	General
97	General
98	General
99	General
100	General

Name	FB_ModeManager	Number	1	Type	FB
Language	LAD	Numbering	Automatic		

Information

Title	Machine Modes and Safety Interlocking Manager	Author	HoraEmbedded	Comment	Monitors physical safety devices and panel inputs to manage operational states. Implements Normally Closed (NC) healthy monitoring for the Emergency Stop button, manages the manual safety fault latch (SR flip-flop), and handles the cycle stop requests (StopPending logic). It ensures a clean segregation between AutoMode and ManualMode to prevent overlapping operational commands.
Family		Version	0.1	User-defined ID	

[illegible]

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Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access-ible from HMI/OPC UA/Web API	Wri-ta-ble from HMI/OPC UA/Web API	Visible in HMI engineering	Set-point	Su-per-vi-sion			Comment
o_Run	Bool	false	Non-retain	True	True	True	False				
o_Fault	Bool	false	Non-retain	True	True	True	False				
o_StopPending	Bool	false	Non-retain	True	True	True	False				
o_AutoMode	Bool	false	Non-retain	True	True	True	False				
o_ManualMode	Bool	false	Non-retain	True	True	True	False				
InOut											
▼ Static											
s_EdgeStart	Bool	false	Non-retain	True	True	True	False				Edge memory bits
s_EdgeReset	Bool	false	Non-retain	True	True	True	False				Edge memory bits
▼ Temp											
t_StartPulse	Bool										One scan pulses
t_ResetPulse	Bool										One scan pulses
Constant											

Network 1: Auto mode

#i_Auto

#i_Manual

#o_AutoMode

Network 2: Manual mode

#i_Manual

#i_Auto

#o_ManualMode

Network 3: Start pulse

#i_Start

P_TRIG

CLK

#s_EdgeStart

Q

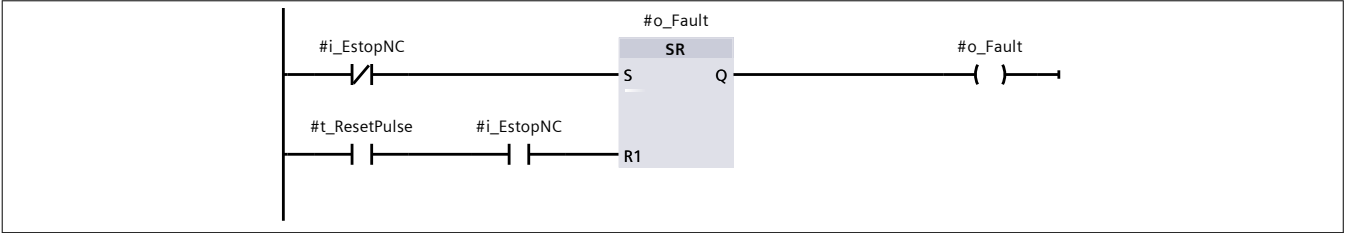
#t_StartPulse

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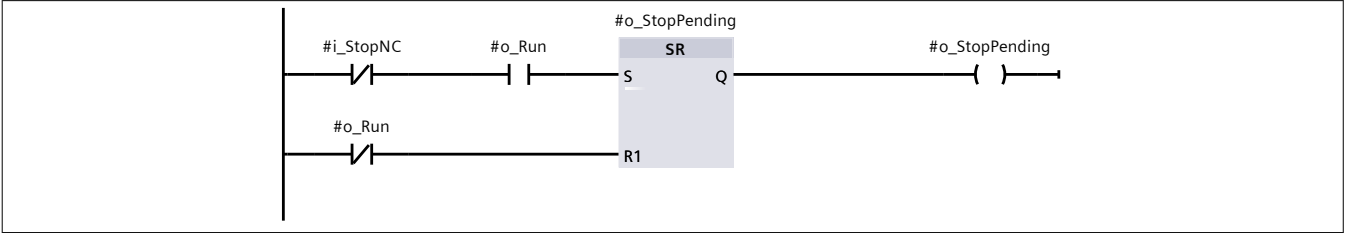
Network 4: Reset pulse



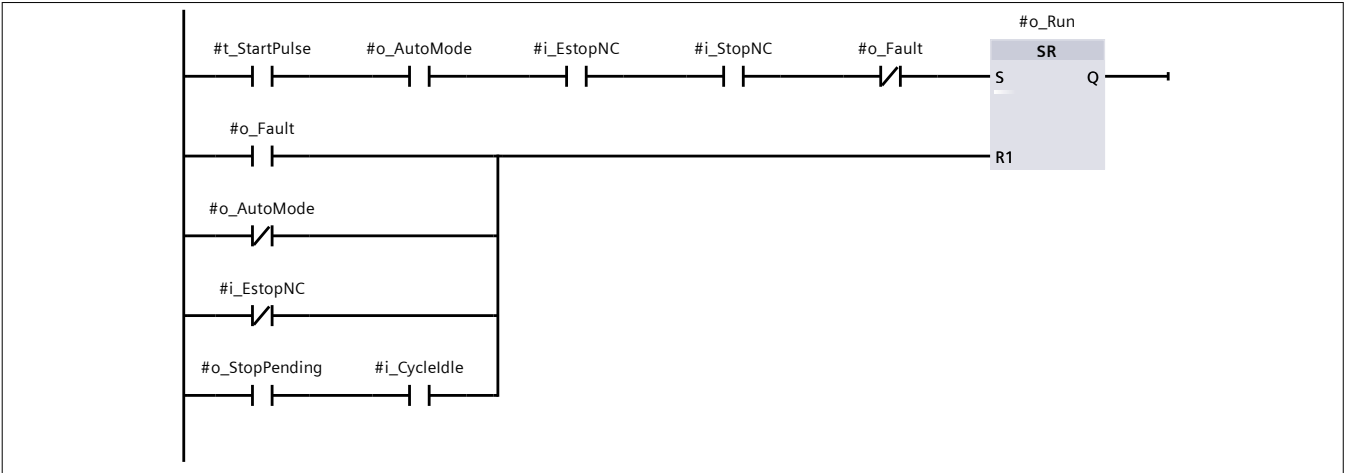
Network 5: Fault latch



Network 6: Stop pending



Network 7: Run latch



SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks

FB_SortingLogic [FB2]

FB_SortingLogic Properties

General	
1	General
2	General
3	General
4	General
5	General
6	General
7	General
8	General
9	General
10	General
11	General
12	General
13	General
14	General
15	General
16	General
17	General
18	General
19	General
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84	General
85	General
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89	General
90	General
91	General
92	General
93	General
94	General
95	General
96	General
97	General
98	General
99	General
100	General

Name	FB_SortingLogic	Number	2	Type	FB
Language	LAD	Numbering	Automatic		

Information

Title	Pipelined Box Sorting Sequence (State Machine)	Author	HoraEmbedded	Comment	Implements the continuous automatic sorting sequence using a customized sequential state machine (Steps 0 to 5). Manages entry box generation, calculates sensor filtration via an independent centering timer block, tracks multiple parts in transit using dedicated software registers (In-Transit counters), and automatically routes boxes based on height measurements recorded by the light curtain.
Family		Version	0.1	User-defined ID	

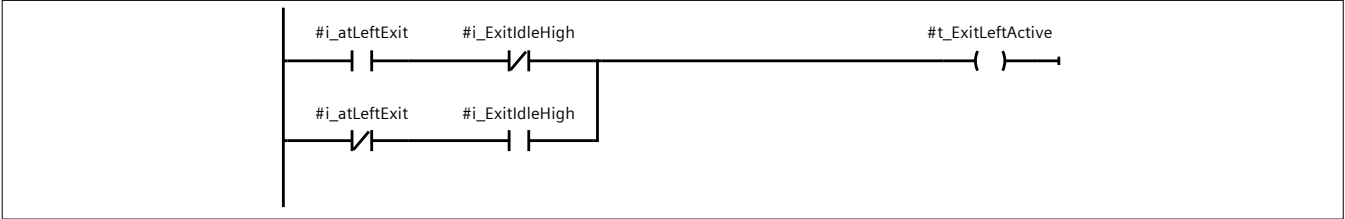
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment
▼ Input									
i_Run	Bool	false	Non-retain	True	True	True	False		
i_StopPending	Bool	false	Non-retain	True	True	True	False		
i_ClearCounters	Bool	false	Non-retain	True	True	True	False		
i_ExitIdleHigh	Bool	false	Non-retain	True	True	True	False		
i_atEntry	Bool	false	Non-retain	True	True	True	False		
i_atLoadPosition	Bool	false	Non-retain	True	True	True	False		
i_atTurntableEntry	Bool	false	Non-retain	True	True	True	False		
i_atUnloadPosition	Bool	false	Non-retain	True	True	True	False		

--	--	--

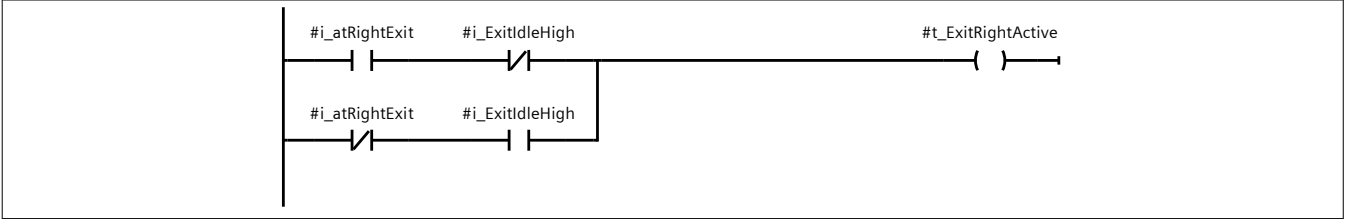
Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion			Comment
i_HighBox	Bool	false	Non-retain	True	True	True	False				
i_atLeftEntry	Bool	false	Non-retain	True	True	True	False				
i_atRightEntry	Bool	false	Non-retain	True	True	True	False				
i_atLeftExit	Bool	false	Non-retain	True	True	True	False				
i_atRightExit	Bool	false	Non-retain	True	True	True	False				
i_CenterTime	Time	T#0ms	Non-retain	True	True	True	False				
i_atFront	Bool	false	Non-retain	True	True	True	False				
▼ Output											
o_CycleIdle	Bool	false	Non-retain	True	True	True	False				
o_Step	DInt	0	Non-retain	True	True	True	False				
▼ InOut											
▼ io_Cmd	"UDT_ActuatorCmd"			False	False	False	False				
EntryConveyor	Bool			False	False	False	False				
FeederConveyor	Bool			False	False	False	False				
LeftConveyor	Bool			False	False	False	False				
RightConveyor	Bool			False	False	False	False				
Load	Bool			False	False	False	False				
Unload	Bool			False	False	False	False				
Turn	Bool			False	False	False	False				
Emit	Bool			False	False	False	False				
RemoverLeft	Bool			False	False	False	False				
RemoverRight	Bool			False	False	False	False				
io_CountLeft	DInt	0	Non-retain	True	True	True	False				
io_CountRight	DInt	0	Non-retain	True	True	True	False				
io_CountTotal	DInt	0	Non-retain	True	True	True	False				

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access-ible from HMI/OPC UA/Web API	Wri-ta-ble from HMI I/O PC UA/ Web API	Visible in HMI engineering	Set-point	Su-per-vi-sion			Comment
▼ Static											
s_Step	DInt	0	Non-retain	True	True	True	False				
s_SeenHigh	Bool	false	Non-retain	True	True	True	False				
s_EdgeExitLeft	Bool	false	Non-retain	True	True	True	False				
s_EdgeExitRight	Bool	false	Non-retain	True	True	True	False				
▼ s_TonEmit	IEC_TIMER		Non-retain	True	True	True	False				
PT	Time	T#0ms	Non-retain	True	True	True	False				
ET	Time	T#0ms	Non-retain	True	False	True	False				
IN	Bool	false	Non-retain	True	True	True	False				
Q	Bool	false	Non-retain	True	False	True	False				
▼ s_TonCenter	IEC_TIMER		Non-retain	True	True	True	True				
PT	Time	T#0ms	Non-retain	True	True	True	False				
ET	Time	T#0ms	Non-retain	True	False	True	False				
IN	Bool	false	Non-retain	True	True	True	False				
Q	Bool	false	Non-retain	True	False	True	False				
s_atEntrySeen	Bool	false	Non-retain	True	True	True	True				
s_LeftInTransit	DInt	0	Non-retain	True	True	True	False				
s_RightInTransit	DInt	0	Non-retain	True	True	True	False				
s_EdgeLeftEntry	Bool	false	Non-retain	True	True	True	False				
s_EdgeRightEntry	Bool	false	Non-retain	True	True	True	False				
▼ Temp											
t_ExitLeftActive	Bool										
t_ExitRightActive	Bool										
t_ExitLeftPulse	Bool										
t_ExitRightPulse	Bool										
t_ChosenEntry	Bool										
t_LeftEntryPulse	Bool										
t_RightEntryPulse	Bool										
Constant											

Network 1: exit left active



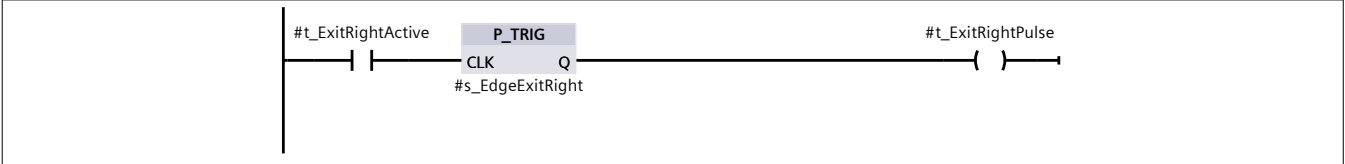
Network 2: exit right active



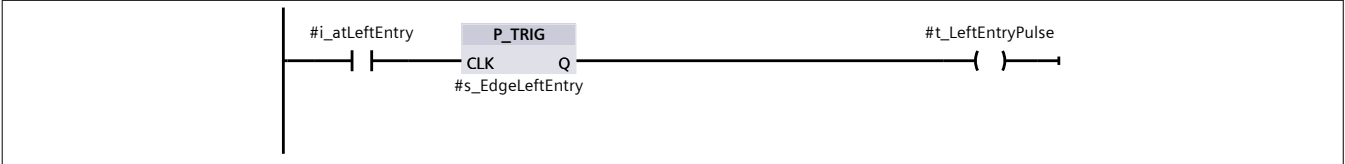
Network 3: exit left pulse



Network 4: exit right pulse



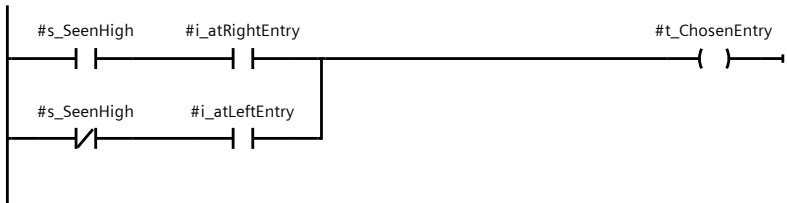
Network 5: Left entry pulse



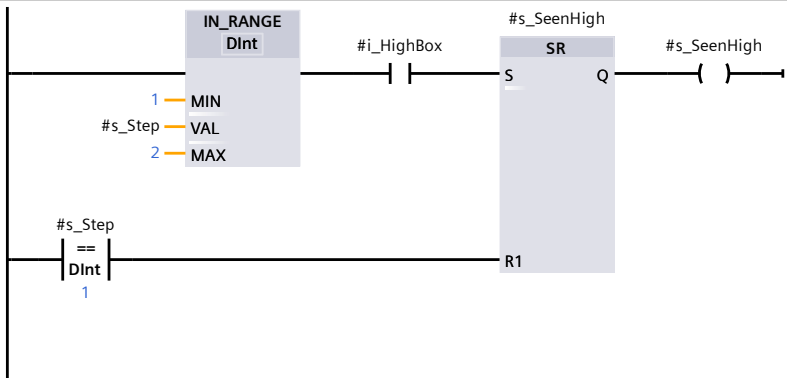
Network 6:



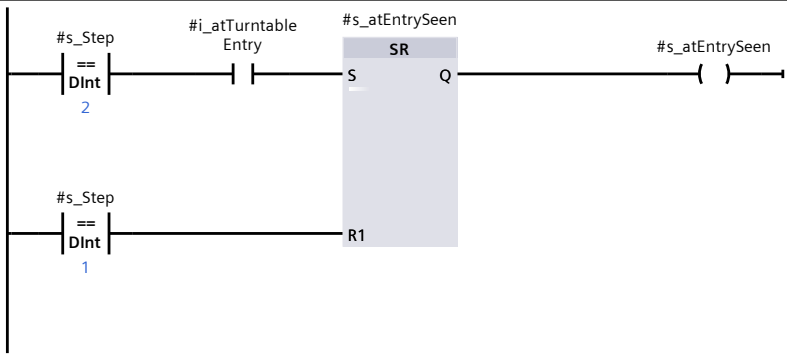
Network 7: chosen entry sensor



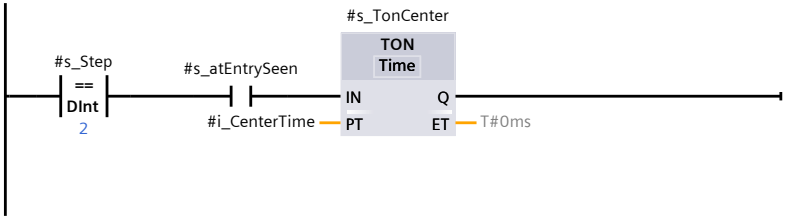
Network 8: height latch



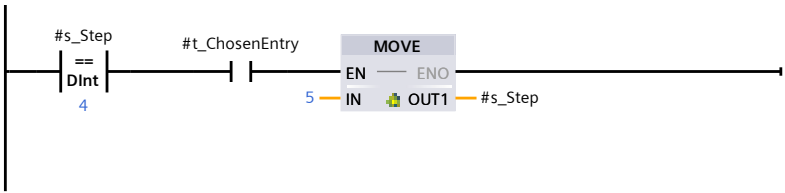
Network 9: Entry seen latch



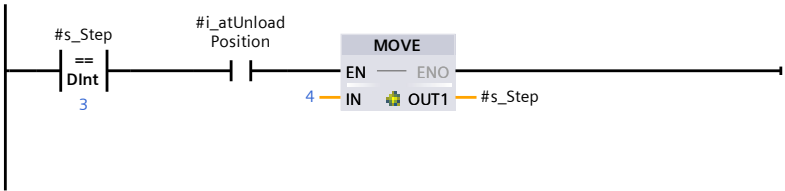
Network 10: Center timer



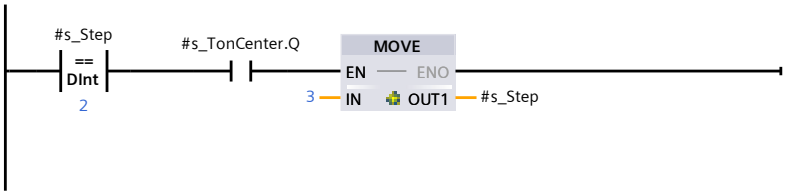
Network 11: T4 to 5 (#s_Step := 5)



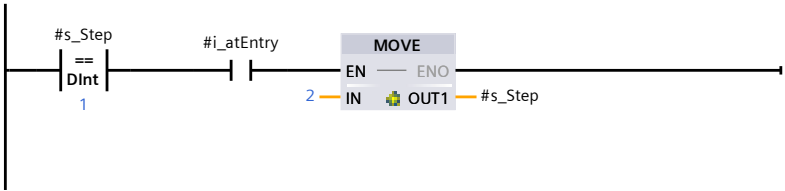
Network 12: T3 to 4 (#s_Step := 4)



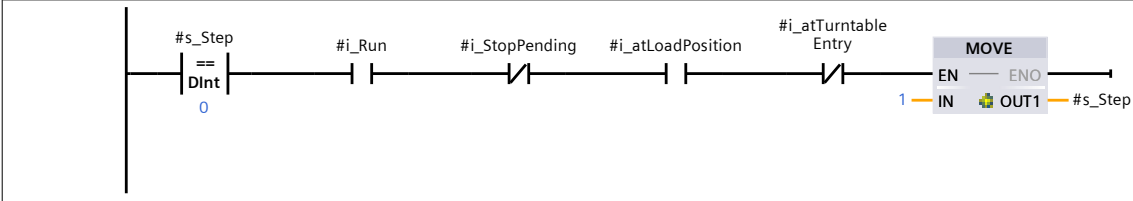
Network 13: T2 to 3 (#s_Step := 3)



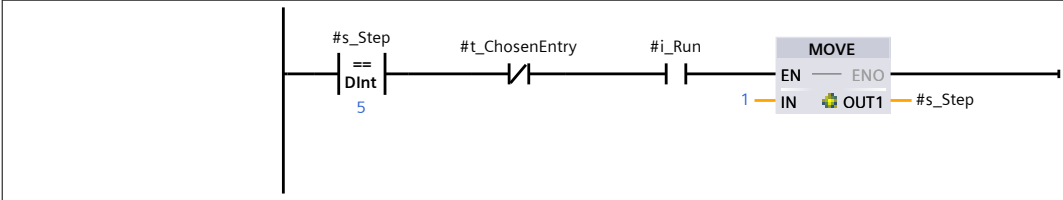
Network 14: T1 to 2 (#s_Step := 2)



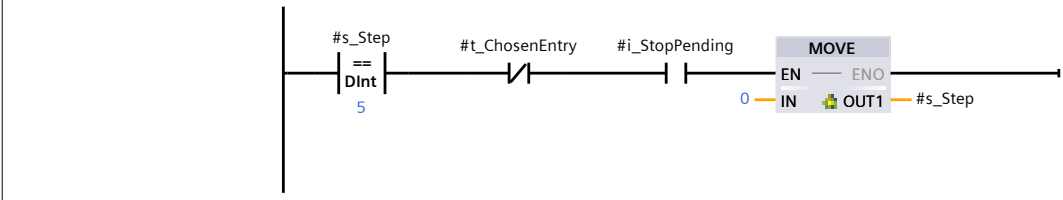
Network 15: T0 to 1 (#s_Step := 1)



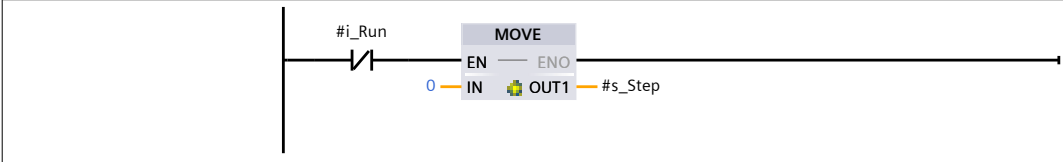
Network 16:



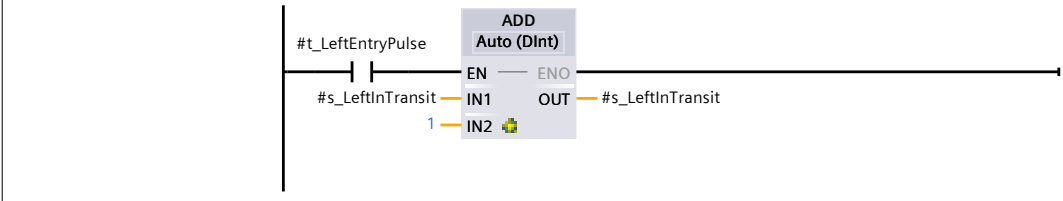
Network 17:



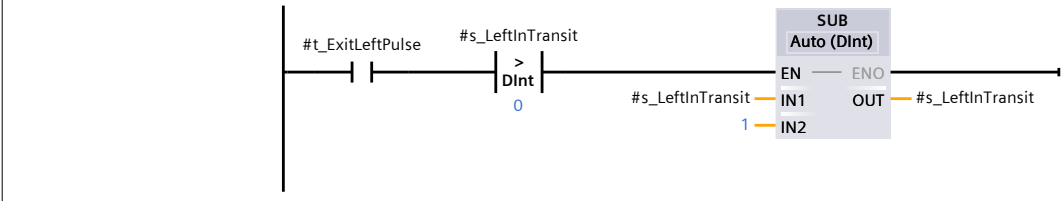
Network 18: Abort when not running



Network 19: Left in transit plus



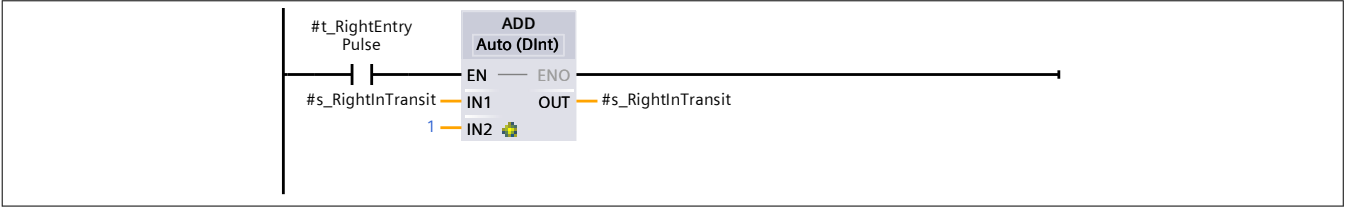
Network 20: Left in transit minus



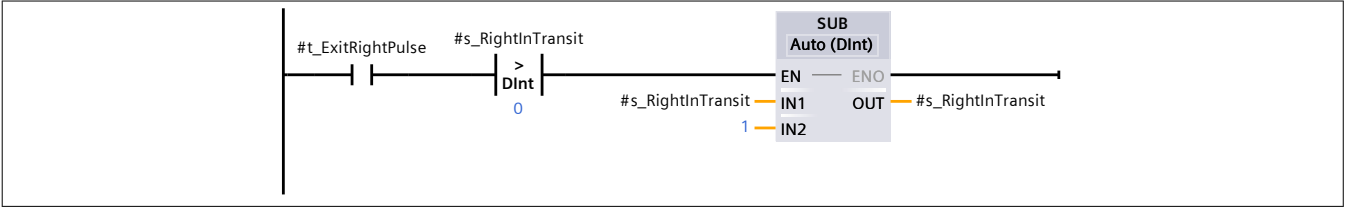
Network 21: Left in transit clear



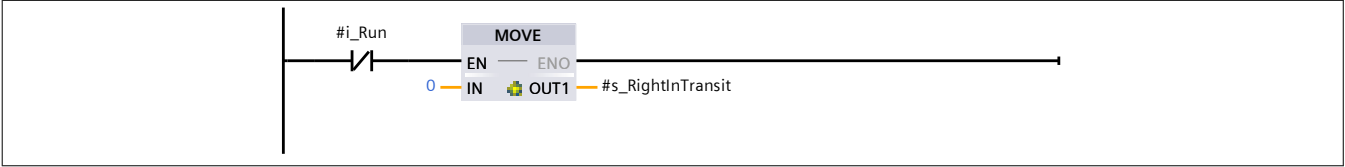
Network 22: Right in transit plus



Network 23: Right in transit minus



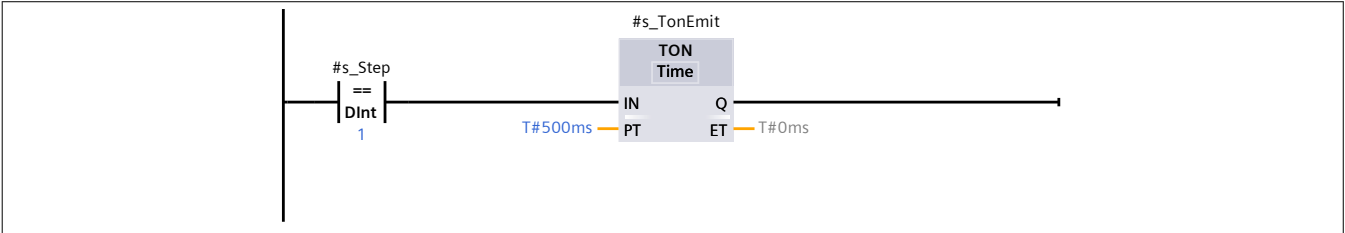
Network 24: Right in transit clear



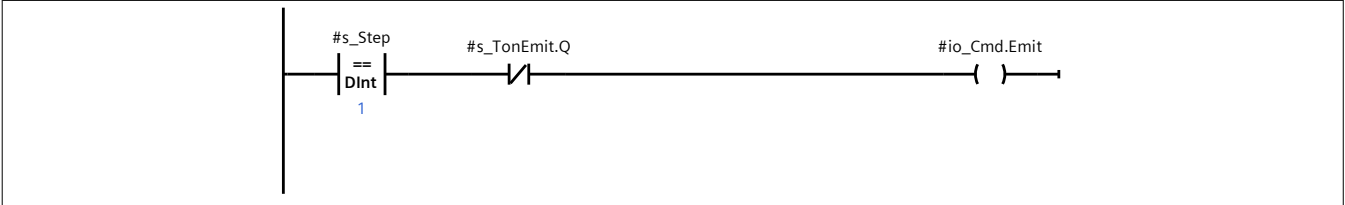
Network 25: Continuous cycle

Network 26: Stop at end of cycle

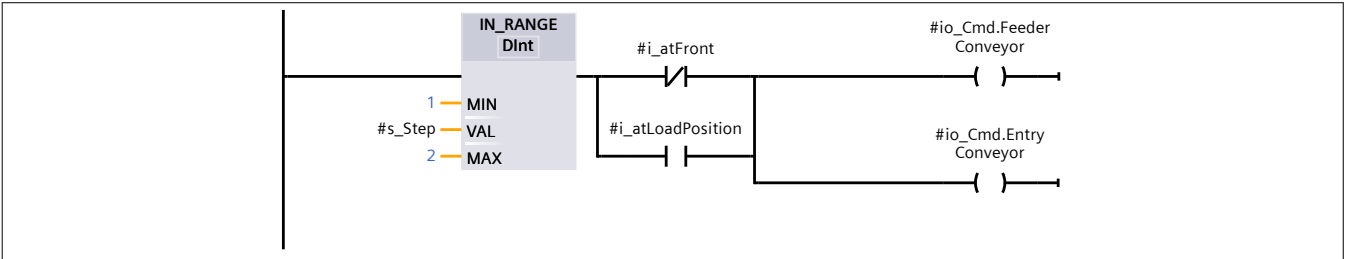
Network 27: Emit timer



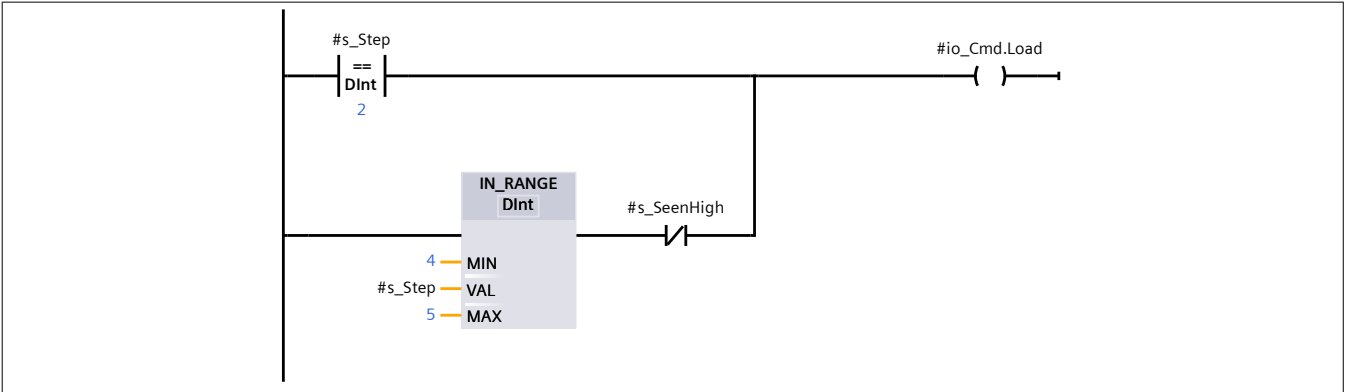
Network 28: Emit command



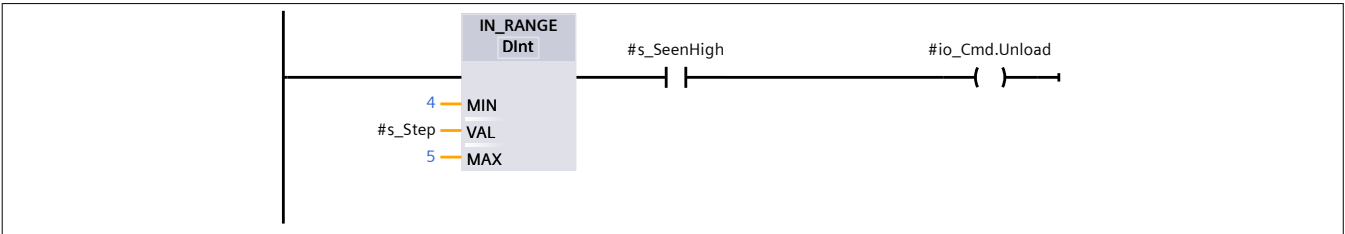
Network 29: Feeder and entry conveyors



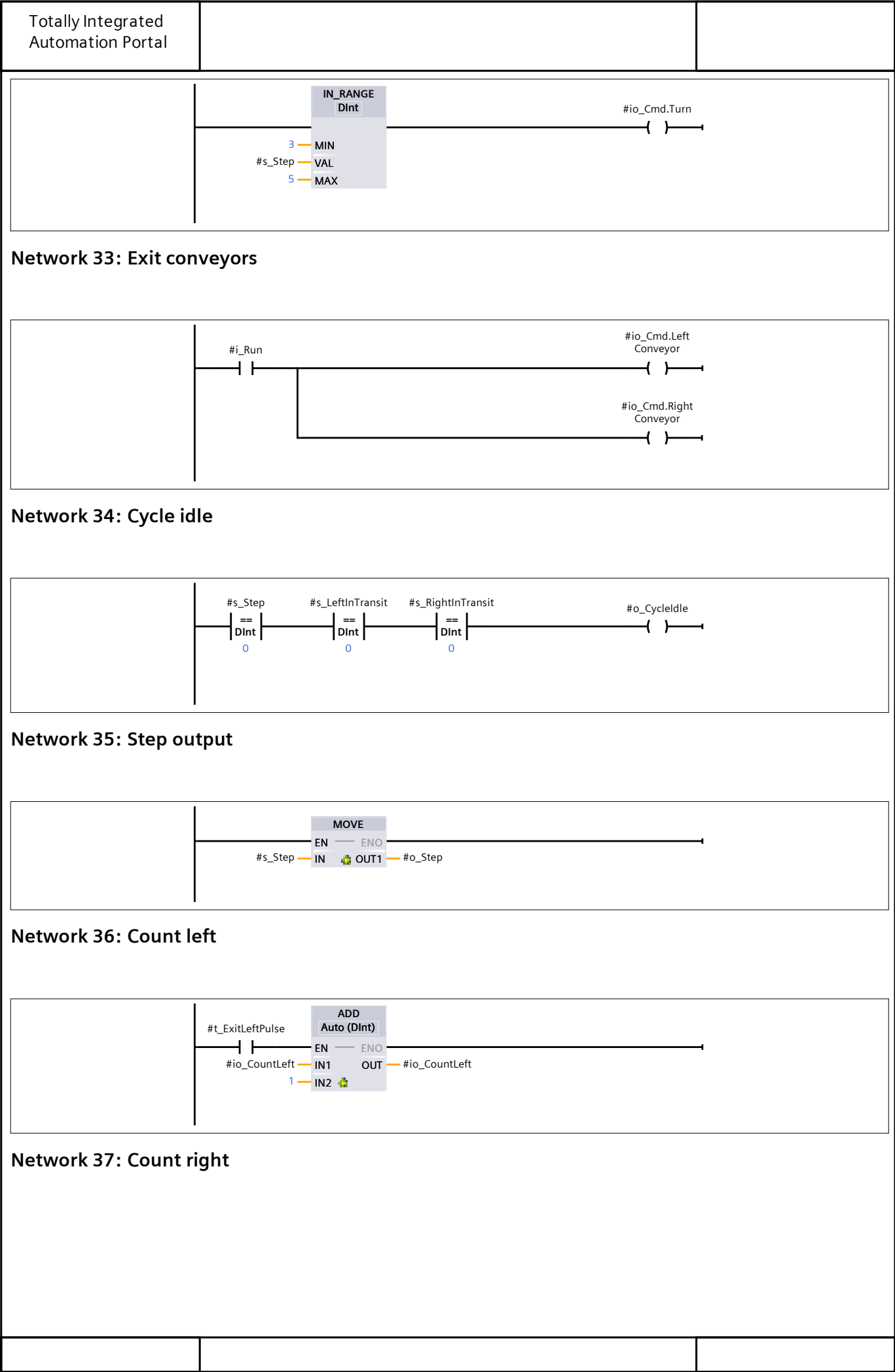
Network 30: Load command

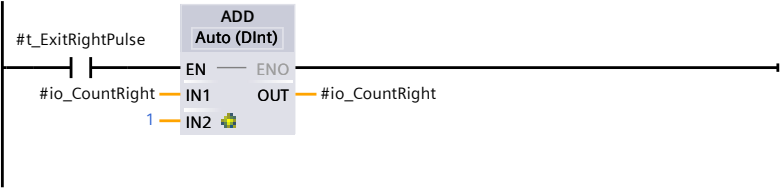


Network 31: Unload command

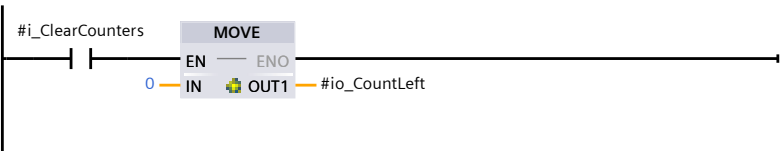


Network 32: Turn command

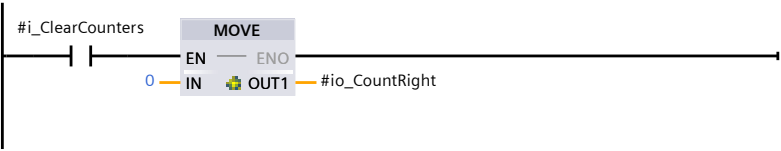




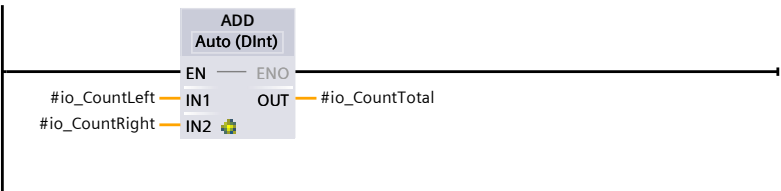
Network 38: Clear left counter



Network 39: Clear right counter



Network 40: Total count



SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Program blocks

FC_outputs [FC1]

FC_outputs Properties

General

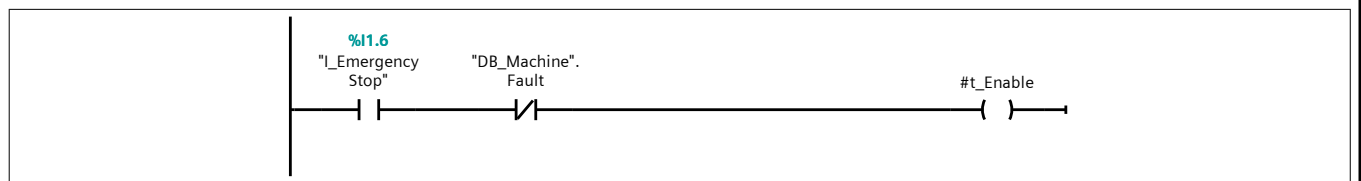
Name	FC_outputs	Number	1	Type	FC
Language	LAD	Numbering	Automatic		

Information

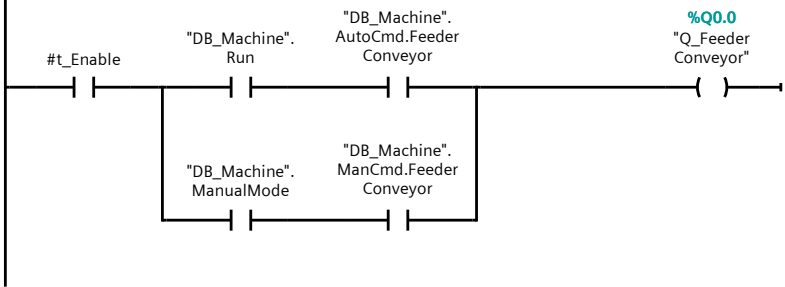
Title	Centralized Output Gating and Hardware Interlocks	Author	HoraEmbedded	Comment	Serves as the single source of truth for all physical digital and analog outputs directed to the Factory I/O simulation panel. It gathers automatic commands generated by the state machine and manual commands from the watch table. It applies strict hardware safety interlocks (System Enable bit derived from the Emergency Stop state) before forcing any physical coil to prevent duplicate assignment conflicts.
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Comment
Input			
Output			
InOut			
▼ Temp			
t_Enable	Bool		
Constant			
▼ Return			
FC_outputs	Void		

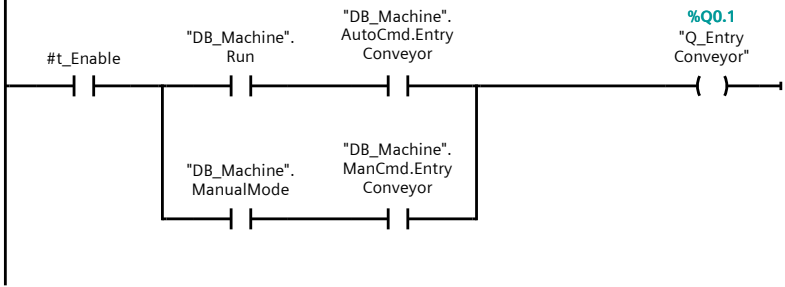
Network 1: Enable



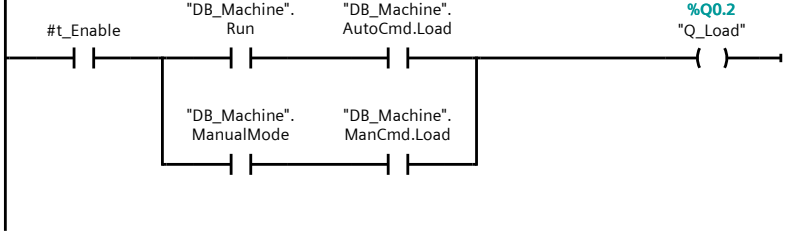
Network 2: feeder conveyor config



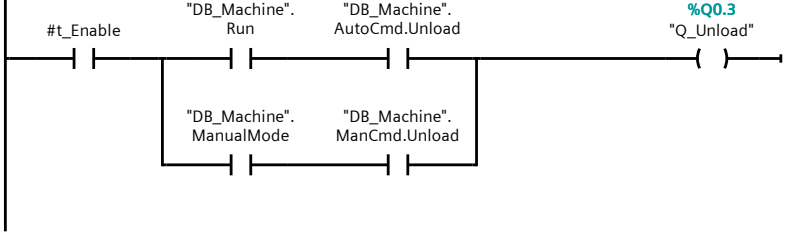
Network 3: entry conveyor config



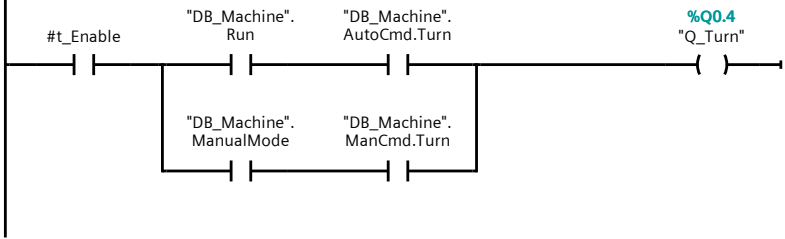
Network 4: Load config



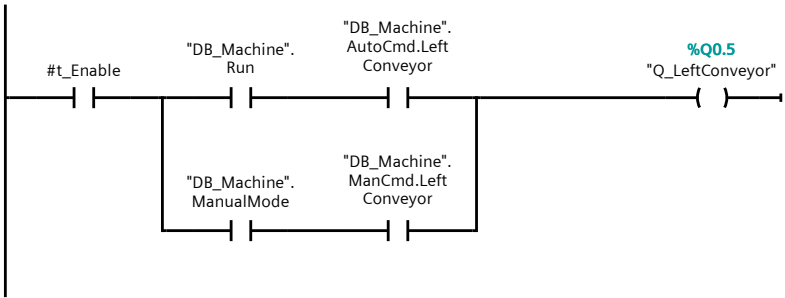
Network 5: Unload config



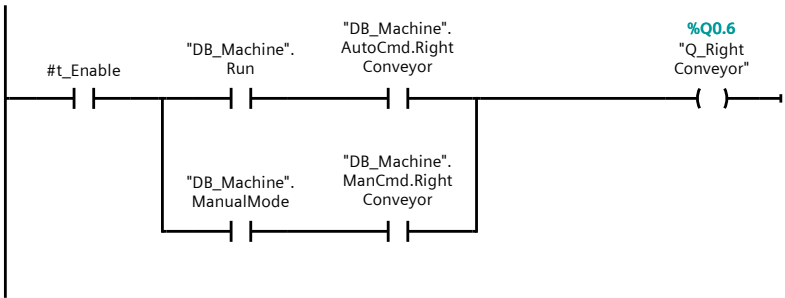
Network 6: Tunr config



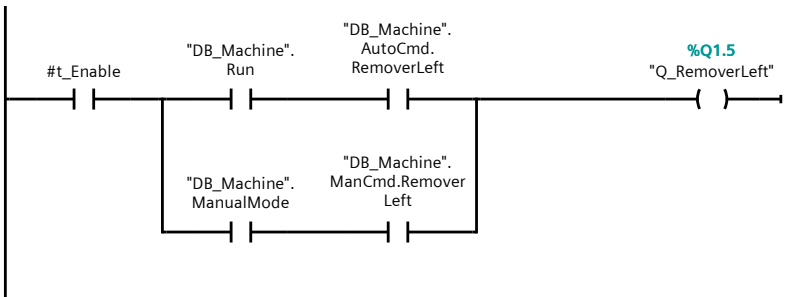
Network 7: Left conveyor config



Network 8: Right conveyor config



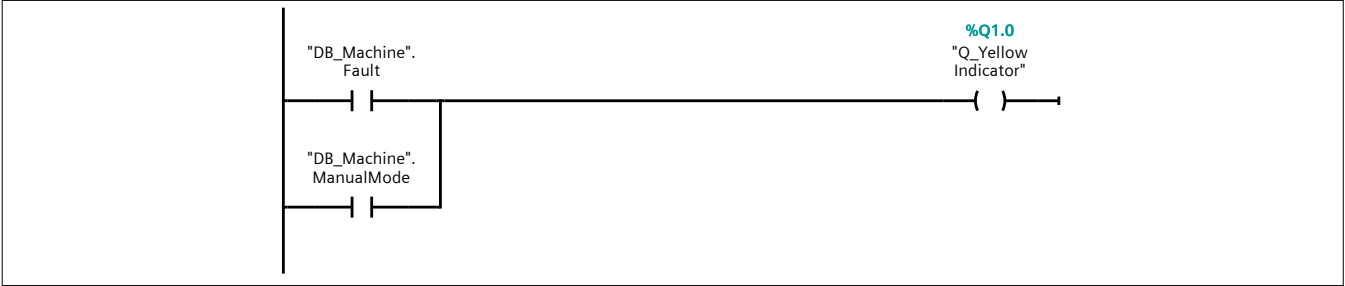
Network 9: Remover left config



Network 10: Remover right config

Totally Integrated Automation Portal		
#t_Enable "DB_Machine".Run "DB_Machine".AutoCmd. RemoverRight "DB_Machine".ManualMode "DB_Machine".ManCmd.Remover Right %Q1.6 "Q_Remover Right"		
Network 11: Emitter config #t_Enable "DB_Machine".Run "DB_Machine".AutoCmd.Emit "DB_Machine".ManualMode "DB_Machine".ManCmd.Emit %Q1.7 "Q_Emit"		
Network 12: Green indicator "DB_Machine".Run %Q0.7 "Q_Green Indicator"		
Network 13: Start light "DB_Machine".Run %Q1.2 "Q_StartLight"		
Network 14: Red indicator "DB_Machine".Run %Q1.1 "Q_RedIndicator"		
Network 15: Stop light "DB_Machine".Run %Q1.4 "Q_StopLight"		

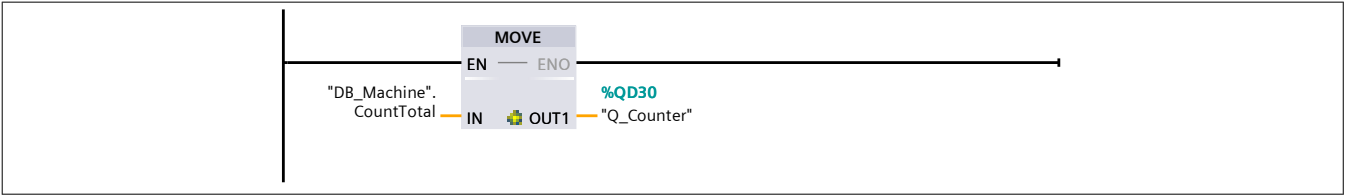
Network 16: Yellow indicator



Network 17: Reset light



Network 18: Counter display



SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks

FB_ModeManager_DB [DB2]

FB_ModeManager_DB Properties

General

Name	FB_ModeManager_DB	Number	2	Type	DB
Language	DB	Numbering	Automatic		

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain	Access- ible from HMI/O PC UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion	Comment
▼ Input									
i_Start	Bool	false	False	True	True	True	False		Start pushbutton
i_StopNC	Bool	false	False	True	True	True	False		Stop pushbutton
i_EstopNC	Bool	false	False	True	True	True	False		Emergency stop
i_Reset	Bool	false	False	True	True	True	False		Reset pushbutton
i_Auto	Bool	false	False	True	True	True	False		Selector in Auto
i_Manual	Bool	false	False	True	True	True	False		Selector in Manual
i_CycleIdle	Bool	false	False	True	True	True	False		No box being processed
▼ Output									
o_Run	Bool	false	False	True	True	True	False		
o_Fault	Bool	false	False	True	True	True	False		
o_StopPending	Bool	false	False	True	True	True	False		
o_AutoMode	Bool	false	False	True	True	True	False		
o_ManualMode	Bool	false	False	True	True	True	False		
InOut									
▼ Static									
s_EdgeStart	Bool	false	False	True	True	True	False		Edge memory bits
s_EdgeReset	Bool	false	False	True	True	True	False		Edge memory bits

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks

FB_SortingLogic_DB [DB3]

FB_SortingLogic_DB Properties

General	
1	General
2	General
3	General
4	General
5	General
6	General
7	General
8	General
9	General
10	General
11	General
12	General
13	General
14	General
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91	General
92	General
93	General
94	General
95	General
96	General
97	General
98	General
99	General
100	General

Name	FB_SortingLogic_DB	Number	3	Type	DB
Language	DB	Numbering	Automatic		

Information	
1	Information
2	Information
3	Information
4	Information
5	Information
6	Information
7	Information
8	Information
9	Information
10	Information
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96	Information
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98	Information
99	Information
100	Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Start value	Retain	Access- ible from HMI/O PC UA/Web API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment
▼ Input									
i_Run	Bool	false	False	True	True	True	False		
i_StopPending	Bool	false	False	True	True	True	False		
i_ClearCounters	Bool	false	False	True	True	True	False		
i_ExitIdleHigh	Bool	false	False	True	True	True	False		
i_atEntry	Bool	false	False	True	True	True	False		
i_atLoadPosition	Bool	false	False	True	True	True	False		
i_atTurntableEntry	Bool	false	False	True	True	True	False		
i_atUnloadPosition	Bool	false	False	True	True	True	False		
i_HighBox	Bool	false	False	True	True	True	False		
i_atLeftEntry	Bool	false	False	True	True	True	False		
i_atRightEntry	Bool	false	False	True	True	True	False		
i_atLeftExit	Bool	false	False	True	True	True	False		
i_atRightExit	Bool	false	False	True	True	True	False		
i_CenterTime	Time	T#0ms	False	True	True	True	False		
i_atFront	Bool	false	False	True	True	True	False		
▼ Output									

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Totally Integrated Automation Portal										
Name	Data type	Start value	Retain	Access- ible from HMI/O PC UA/Web API	Wri- ta- ble from HMI/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Sup- er- vi- sion	Comment	
o_CycleIdle	Bool	false	False	True	True	True	False			
o_Step	DInt	0	False	True	True	True	False			
▼ InOut										
io_Cmd	"UDT_ActuatorCmd"		False	False	False	False	False			
io_CountLeft	DInt	0	False	True	True	True	False			
io_CountRight	DInt	0	False	True	True	True	False			
io_CountTotal	DInt	0	False	True	True	True	False			
▼ Static										
s_Step	DInt	0	False	True	True	True	False			
s_SeenHigh	Bool	false	False	True	True	True	False			
s_EdgeExitLeft	Bool	false	False	True	True	True	False			
s_EdgeExitRight	Bool	false	False	True	True	True	False			
▼ s_TonEmit	IEC_TIMER		False	True	True	True	False			
PT	Time	T#0ms	False	True	True	True	False			
ET	Time	T#0ms	False	True	False	True	False			
IN	Bool	false	False	True	True	True	False			
Q	Bool	false	False	True	False	True	False			
▼ s_TonCenter	IEC_TIMER		False	True	True	True	True			
PT	Time	T#0ms	False	True	True	True	False			
ET	Time	T#0ms	False	True	False	True	False			
IN	Bool	false	False	True	True	True	False			
Q	Bool	false	False	True	False	True	False			
s_atEntrySeen	Bool	false	False	True	True	True	True			
s_LeftInTransit	DInt	0	False	True	True	True	False			
s_RightInTransit	DInt	0	False	True	True	True	False			

**SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Program blocks / System blocks**

Program resources

This folder is empty.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC]

Technology objects

This folder is empty.

[illegible]

PLC tags

PLC tags									
	Name	Data type	Address	Retain	Access- ible from HMI/O PC UA/W eb API	Writa- ble from HMI/O PC UA/W eb API	Visi- ble in HMI engi- neer- ing	Supervision	Comment

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC tags /
Default tag table [29]

User constants

User constants			
Name	Data type	Value	Comment

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC tags / Factory_IO_Inputs [18]

PLC tags

PLC tags											
	Name	Data type	Address	Retain	Access- ible from HMI/O PC UA/W eb API	Writa- ble from HMI/O PC UA/W eb API	Visi- ble in HMI engi- neer- ing	Supervision			Comment
	I_atBack	Bool	%I2.1	False	True	True	True				Confirms a box has fully cleared the entry area.
	I_atEntry	Bool	%I0.0	False	True	True	True				Detects box emergence.
	I_atFront	Bool	%I0.6	False	True	True	True				Prevents part collisions.
	I_atLeftEntry	Bool	%I1.0	False	True	True	True				Confirms the box has successfully transitioned onto the Left exit convey- or.
	I_atLeftExit	Bool	%I1.2	False	True	True	True				Drops to FALSE when a box reaches the left re- mover.
	I_atLoadPosition	Bool	%I0.4	False	True	True	True				TRUE when turntable is at 0° (aligned with en- try).
	I_atRightEntry	Bool	%I0.7	False	True	True	True				Confirms the box has successfully transitioned onto the Right exit con- veyor.
	I_atRightExit	Bool	%I1.1	False	True	True	True				Drops to FALSE when a box reaches the right re- mover.
	I_atTurntableEntry	Bool	%I0.3	False	True	True	True				Center reflective sensor inside the turntable. Used for precise stop- ping.
	I_atUnloadPosition	Bool	%I0.5	False	True	True	True				TRUE when turntable has completed its 90° clock- wise rotation.
	I_Auto	Bool	%I1.7	False	True	True	True				Selector in Auto position.
	I_Manual	Bool	%I2.2	False	True	True	True				Selector in Manual posi- tion.
	I_EmergencyStop	Bool	%I1.6	False	True	True	True				TRUE when healthy, drops to FALSE when pressed.
	I_LowBox	Bool	%I0.1	False	True	True	True				Active (TRUE) for both low and high boxes.
	I_Reset	Bool	%I1.4	False	True	True	True				Clears safety fault latch- es.
	I_Start	Bool	%I1.3	False	True	True	True				Triggers the cycle run latch.
	I_Stop	Bool	%I1.5	False	True	True	True				TRUE at rest, drops to FALSE to request a cycle stop.
	I_HighBox	Bool	%I0.2	False	True	True	True				Active (TRUE) only along- side LowBox for high boxes.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC tags /
Factory_IO_Inputs [18]

User constants

User constants			
Name	Data type	Value	Comment

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC tags / Factory_IO_Outputs [17]

PLC tags

PLC tags											
	Name	Data type	Address	Retain	Access- ible from HMI/O PC UA/W eb API	Writa- ble from HMI/O PC UA/W eb API	Visi- ble in HMI engi- neer- ing	Supervision			Comment
	Q_Counter	DInt	%QD30	False	True	True	True				Digital totalizer display unit located on the operator panel.
	Q_EntryConveyor	Bool	%Q0.1	False	True	True	True				Drives the second section of the entry lane towards the turntable.
	Q_FeederConveyor	Bool	%Q0.0	False	True	True	True				Drives the first entry section right beneath the box emitter.
	Q_GreenIndicator	Bool	%Q0.7	False	True	True	True				Constant green indicates system is running active cycles.
	Q_LeftConveyor	Bool	%Q0.5	False	True	True	True				Drives the left clearance exit lane.
	Q_Load	Bool	%Q0.2	False	True	True	True				Runs internal turntable rollers forward to pull a box inside.
	Q_RedIndicator	Bool	%Q1.1	False	True	True	True				Constant red indicates system is idle/stopped.
	Q_ResetLight	Bool	%Q1.3	False	True	True	True				Active when safety is tripped and waiting for Reset.
	Q_RightConveyor	Bool	%Q0.6	False	True	True	True				Drives the right clearance exit lane.
	Q_StartLight	Bool	%Q1.2	False	True	True	True				Follows the Run state.
	Q_StopLight	Bool	%Q1.4	False	True	True	True				Active when cycle is off-line.
	Q_Turn	Bool	%Q0.4	False	True	True	True				Swivels the turntable mechanism 90° clockwise.
	Q_Unload	Bool	%Q0.3	False	True	True	True				Runs internal turntable rollers backward to eject a box out to the left lane.
	Q_YellowIndicator	Bool	%Q1.0	False	True	True	True				Active on safety Fault or when system is in Manual mode.
	Q_RemoverLeft	Bool	%Q1.5	False	True	True	True				Instantly deletes boxes at the left exit limit.
	Q_RemoverRight	Bool	%Q1.6	False	True	True	True				Instantly deletes boxes at the right exit limit.
	Q_Emit	Bool	%Q1.7	False	True	True	True				Continuous TRUE spawns boxes sequentially.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC tags /
Factory_IO_Outputs [17]

User constants

User constants			
Name	Data type	Value	Comment

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
PLC data types

UDT_ActuatorCmd

UDT_ActuatorCmd Properties					
General					
Name	UDT_ActuatorCmd	Number	1	Type	UDT
Language		Numbering			
Information					
Title		Author		Comment	
Family		Version		User-defined ID	

Name	Data type	Default value	Accessi-ble from HMI/OP C UA/Web API	Wri-ta-ble from HMI I/O PC UA/We b API	Visi-ble in HMI engi-neer-ing	Set-point	Su-per-vi-sion	Comment
EntryConveyor	Bool	false	True	True	True	False		
FeederConveyor	Bool	false	True	True	True	False		
LeftConveyor	Bool	false	True	True	True	False		
RightConveyor	Bool	false	True	True	True	False		
Load	Bool	false	True	True	True	False		
Unload	Bool	false	True	True	True	False		
Turn	Bool	false	True	True	True	False		
Emit	Bool	false	True	True	True	False		
RemoverLeft	Bool	true	True	True	True	False		
RemoverRight	Bool	true	True	True	True	False		

Totally Integrated Automation Portal		
SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / PLC data types System data types This folder is empty.		

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Watch and force tables

Force table

Name	Address	Display format	Force value	Comment
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SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Watch and force tables

WT_Manual

Name	Address	Display format	Modify value	Comment
"DB_Ma- chine".ManCmd.Feeder- Conveyor		Bool		
"DB_Ma- chine".ManCmd.Entry- Conveyor		Bool		
"DB_Ma- chine".ManCmd.Load		Bool		
"DB_Ma- chine".ManCmd.Unload		Bool		
"DB_Ma- chine".ManCmd.Turn		Bool		
"DB_Ma- chine".ManCmd.Left- Conveyor		Bool		
"DB_Ma- chine".ManCmd.Right- Conveyor		Bool		
"DB_Ma- chine".ManCmd.Emit		Bool		
"DB_Ma- chine".ManCmd.Remov- erLeft		Bool		
"DB_Ma- chine".ManCmd.Remov- erRight		Bool		
"I_atEntry"	%I0.0	Bool		
"I_LowBox"	%I0.1	Bool		
"I_HighBox"	%I0.2	Bool		
"I_atFront"	%I0.6	Bool		
"I_atTurntableEntry"	%I0.3	Bool		
"I_atLoadPosition"	%I0.4	Bool		
"I_atUnloadPosition"	%I0.5	Bool		
"I_atLeftEntry"	%I1.0	Bool		
"I_atRightExit"	%I1.1	Bool		
"I_atBack"	%I2.1	Bool		
"DB_Machine".CountTo- tal		DEC+/-		
"Q_Counter"	%QD30	DEC+/-		

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Watch and force tables

WT_Mode

Name	Address	Display format	Modify value	Comment
"I_Start"	%I1.3	Bool		
"I_Stop"	%I1.5	Bool		
"I_EmergencyStop"	%I1.6	Bool		
"I_Reset"	%I1.4	Bool		
"I_Auto"	%I1.7	Bool		
"I_Manual"	%I2.2	Bool		
"DB_Machine".Run		Bool		
"DB_Machine".Fault		Bool		
"DB_Machine".StopPend- ing		Bool		
"DB_Machine".Auto- Mode		Bool		
"DB_Machine".Manual- Mode		Bool		
"DB_Machine".CycleIdle		Bool		
"Q_YellowIndicator"	%Q1.0	Bool		
"Q_StartLight"	%Q1.2	Bool		
"Q_StopLight"	%Q1.4	Bool		
"Q_ResetLight"	%Q1.3	Bool		
"Q_GreenIndicator"	%Q0.7	Bool		
"Q_RedIndicator"	%Q1.1	Bool		

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Watch and force tables

WT_Sorting

Name	Address	Display format	Modify value	Comment
"Q_Emit"	%Q1.7	Bool		
"Q_FeederConveyor"	%Q0.0	Bool		
"Q_EntryConveyor"	%Q0.1	Bool		
"Q_Load"	%Q0.2	Bool		
"Q_Unload"	%Q0.3	Bool		
"Q_Turn"	%Q0.4	Bool		
"Q_LeftConveyor"	%Q0.5	Bool		
"Q_RightConveyor"	%Q0.6	Bool		
"I_atEntry"	%I0.0	Bool		
"I_atTurntableEntry"	%I0.3	Bool		
"I_atUnloadPosition"	%I0.5	Bool		
"I_atLoadPosition"	%I0.4	Bool		
"I_atLeftEntry"	%I1.0	Bool		
"I_atRightEntry"	%I0.7	Bool		
"I_atLeftExit"	%I1.2	Bool		
"I_atRightExit"	%I1.1	Bool		
"DB_Machine".Run		Bool		
"DB_Machine".SortStep		DEC+/-		
"DB_Machine".StopPend- ing		Bool		
"DB_Machine".CycleIdle		Bool		
"DB_Machine".Fault		Bool		
"DB_Machine".CountTo- tal		DEC+/-		
"DB_Machine".ExitIdle- High		Bool		
"FB_SortingLog- ic_DB".s_atEntrySeen		Bool		
"FB_SortingLog- ic_DB".s_TonCenter.ET		Time		
"FB_SortingLog- ic_DB".s_TonCenter.PT		Time		
"DB_Machine".Center- Time		Time		
"FB_SortingLog- ic_DB".s_SeenHigh		Bool		
"I_LowBox"	%I0.1	Bool		
"I_HighBox"	%I0.2	Bool		
"DB_Machine".Count- Right		DEC+/-		
"DB_Machine".CountLeft		DEC+/-		

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC]

Traces

Name

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Traces
Measurements

This folder is empty.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] / Traces
Combined measurements

Name

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
OPC UA communication

Server interfaces

This folder is empty.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC]

PLC alarm text lists

This folder is empty.

SortingLine_Project / PLC_Sorting [CPU 1214C DC/DC/DC] /
Local modules

PLC_Sorting [CPU 1214C DC/DC/DC]

PLC_Sorting

General\Project information

Name	PLC_Sorting	Author	HoraEmbedded
Comment		Slot	1
Rack	0		

General\Catalog information

Short designation	CPU 1214C DC/DC/DC	Description	Work memory 150 KB; 24VDC power supply with DI14 x 24VDC SINK/ SOURCE, DQ10 x 24VDC and AI2 on board; 6 high-speed counters and 4 pulse outputs on-board; signal board expands on-board I/O; up to 3 communication modules for serial communication; up to 8 signal modules for I/O expansion; PROFINET IO controller, I-device, transport protocol TCP/IP, secure Open User Communication, S7 communication, Web server, OPC UA: Server DA
Article number	6ES7 214-1AG40-0XB0	Firmware version	V4.6
	False		

General\Identification & Maintenance

Plant designation		Location identifier	
Installation date	2026-09-28 22:38:48.935	Additional information	

General\Checksums

Text lists	FA 70 E8 75 1D 5A 8E 29	Software	Not available (compile necessary)
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PROFINET interface [X1]\General

Name	PROFINET interface_1	Author	HoraEmbedded
Comment			

PROFINET interface [X1]\General\Project information

Name	DI 14/DQ 10_1	Comment	
Name	AI 2_1	Comment	

PROFINET interface [X1]\Ethernet addresses\Interface networked with

Subnet:	Not connected		
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PROFINET interface [X1]\Ethernet addresses\Internet protocol version 4 (IPv4)

IP configuration	Set IP address in the project	IP address:	192.168.0.1
Subnet mask:	255.255.255.0	Use router	False

PROFINET interface [X1]\Ethernet addresses\PROFINET

PROFINET device name is set directly at the device	False	Generate PROFINET device name automatically	True
PROFINET device name:	plc_sorting	Converted name:	plcxbsorting9923
Device number:	0		

PROFINET interface [X1]\Time synchronization

Enable time synchronization via NTP server	Enable time synchronization via NTP server		IP addresses
Server 1	0.0.0.0	Server 2	0.0.0.0
Server 3	0.0.0.0	Server 4	0.0.0.0
Update interval	10sec		

Totally Integrated Automation Portal		
CPU synchronizes the modules of the device.	No synchronization	
PROFINET interface [X1]\Digital inputs\Channel0		
Channel address	I100.0	Input filters 6.4 millise
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel0\		
Enable rising edge detection	0	Prefix Event Rising Edge 49152
Event name:	0	Hardware interrupt: 0
Rising edge0	Rising edge0	
PROFINET interface [X1]\Digital inputs\Channel0\		
Enable falling edge detection	0	Prefix Event Falling Edge 49280
Event name:	0	Hardware interrupt: 0
Falling edge0	Falling edge0	
PROFINET interface [X1]\Digital inputs\Channel1		
Channel address	I100.1	Input filters 6.4 millise
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel1\		
Enable rising edge detection	0	Prefix Event Rising Edge 49153
Event name:	0	Hardware interrupt: 0
Rising edge1	Rising edge1	
PROFINET interface [X1]\Digital inputs\Channel1\		
Enable falling edge detection	0	Prefix Event Falling Edge 49281
Event name:	0	Hardware interrupt: 0
Falling edge1	Falling edge1	
PROFINET interface [X1]\Digital inputs\Channel2		
Channel address	I100.2	Input filters 6.4 millise
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel2\		
Enable rising edge detection	0	Prefix Event Rising Edge 49154
Event name:	0	Hardware interrupt: 0
Rising edge2	Rising edge2	
PROFINET interface [X1]\Digital inputs\Channel2\		
Enable falling edge detection	0	Prefix Event Falling Edge 49282
Event name:	0	Hardware interrupt: 0
Falling edge2	Falling edge2	
PROFINET interface [X1]\Digital inputs\Channel3		
Channel address	I100.3	Input filters 6.4 millise
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel3\		
Enable rising edge detection	0	Prefix Event Rising Edge 49155
Event name:	0	Hardware interrupt: 0
Rising edge3	Rising edge3	
PROFINET interface [X1]\Digital inputs\Channel3\		
Enable falling edge detection	0	Prefix Event Falling Edge 49283
Event name:	0	Hardware interrupt: 0
Falling edge3	Falling edge3	
PROFINET interface [X1]\Digital inputs\Channel4		
Channel address	I100.4	Input filters 6.4 millise
Enable pulse catch	0	

Totally Integrated Automation Portal		
PROFINET interface [X1]\Digital inputs\Channel4\		
Enable rising edge detection	0	Prefix Event Rising Edge 49156
Event name:	0	Hardware interrupt: 0
Rising edge4	Rising edge4	
PROFINET interface [X1]\Digital inputs\Channel4\		
Enable falling edge detection	0	Prefix Event Falling Edge 49284
Event name:	0	Hardware interrupt: 0
Falling edge4	Falling edge4	
PROFINET interface [X1]\Digital inputs\Channel5		
Channel address	I100.5	Input filters 6.4 millisec
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel5\		
Enable rising edge detection	0	Prefix Event Rising Edge 49157
Event name:	0	Hardware interrupt: 0
Rising edge5	Rising edge5	
PROFINET interface [X1]\Digital inputs\Channel5\		
Enable falling edge detection	0	Prefix Event Falling Edge 49285
Event name:	0	Hardware interrupt: 0
Falling edge5	Falling edge5	
PROFINET interface [X1]\Digital inputs\Channel6		
Channel address	I100.6	Input filters 6.4 millisec
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel6\		
Enable rising edge detection	0	Prefix Event Rising Edge 49158
Event name:	0	Hardware interrupt: 0
Rising edge6	Rising edge6	
PROFINET interface [X1]\Digital inputs\Channel6\		
Enable falling edge detection	0	Prefix Event Falling Edge 49286
Event name:	0	Hardware interrupt: 0
Falling edge6	Falling edge6	
PROFINET interface [X1]\Digital inputs\Channel7		
Channel address	I100.7	Input filters 6.4 millisec
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel7\		
Enable rising edge detection	0	Prefix Event Rising Edge 49159
Event name:	0	Hardware interrupt: 0
Rising edge7	Rising edge7	
PROFINET interface [X1]\Digital inputs\Channel7\		
Enable falling edge detection	0	Prefix Event Falling Edge 49287
Event name:	0	Hardware interrupt: 0
Falling edge7	Falling edge7	
PROFINET interface [X1]\Digital inputs\Channel8		
Channel address	I101.0	Input filters 6.4 millisec
Enable pulse catch	0	
PROFINET interface [X1]\Digital inputs\Channel8\		
Enable rising edge detection	0	Prefix Event Rising Edge 49160
Event name:	0	Hardware interrupt: 0
Rising edge8	Rising edge8	

Totally Integrated Automation Portal					
PROFINET interface [X1]\Digital inputs\Channel8\					
Enable falling edge detection	0	Prefix Event Falling Edge	49288		
Event name:	0	Hardware interrupt:	0		
Falling edge8	Falling edge8				
PROFINET interface [X1]\Digital inputs\Channel9					
Channel address	I101.1	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel9\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49161		
Event name:	0	Hardware interrupt:	0		
Rising edge9	Rising edge9				
PROFINET interface [X1]\Digital inputs\Channel9\					
Enable falling edge detection	0	Prefix Event Falling Edge	49289		
Event name:	0	Hardware interrupt:	0		
Falling edge9	Falling edge9				
PROFINET interface [X1]\Digital inputs\Channel10					
Channel address	I101.2	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel10\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49162		
Event name:	0	Hardware interrupt:	0		
Rising edge10	Rising edge10				
PROFINET interface [X1]\Digital inputs\Channel10\					
Enable falling edge detection	0	Prefix Event Falling Edge	49290		
Event name:	0	Hardware interrupt:	0		
Falling edge10	Falling edge10				
PROFINET interface [X1]\Digital inputs\Channel11					
Channel address	I101.3	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel11\					
Enable rising edge de- tection	0	Prefix Event Rising Edge	49163		
Event name:	0	Hardware interrupt:	0		
Rising edge11	Rising edge11				
PROFINET interface [X1]\Digital inputs\Channel11\					
Enable falling edge detection	0	Prefix Event Falling Edge	49291		
Event name:	0	Hardware interrupt:	0		
Falling edge11	Falling edge11				
PROFINET interface [X1]\Digital inputs\Channel12					
Channel address	I101.4	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Digital inputs\Channel13					
Channel address	I101.5	Input filters	6.4 millise		
Enable pulse catch	0				
PROFINET interface [X1]\Analog inputs\Noise reduction					
Integration time	50 Hz (20 ms)				
PROFINET interface [X1]\Analog inputs\Channel0					
Channel address	IW64	Measurement type	Voltage		
Voltage range	0..10 V	Smoothing	Weak (4 cycles)		
		Enable overflow diag- nostics	1		
PROFINET interface [X1]\Analog inputs\Channel1					
Channel address	IW66	Measurement type	Voltage		

Totally Integrated Automation Portal					
Voltage range	0..10 V		Smoothing	Weak (4 cycles)	
			Enable overflow diagnostics	1	
PROFINET interface [X1]\Digital outputs					
Reaction to CPU STOP	Use substitute value				
PROFINET interface [X1]\Digital outputs\Channel0					
Channel address	Q0.0		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel1					
Channel address	Q0.1		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel2					
Channel address	Q0.2		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel3					
Channel address	Q0.3		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel4					
Channel address	Q0.4		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel5					
Channel address	Q0.5		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel6					
Channel address	Q0.6		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel7					
Channel address	Q0.7		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel8					
Channel address	Q1.0		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Digital outputs\Channel9					
Channel address	Q1.1		Substitute a value of 1 on a change from RUN to STOP.	0	
PROFINET interface [X1]\Operating mode					
IO controller	True		IO system		
Device number	0		IO device	False	
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	100.0		End address	101.7	
Organization block	0		Process image	0	
PROFINET interface [X1]\I/O addresses\Input addresses					
Start address	64		End address	67	
Organization block	0		Process image	0	
PROFINET interface [X1]\I/O addresses\Output addresses					
Start address	0.0		End address	1.7	
Organization block	0		Process image	0	

Totally Integrated Automation Portal			
PROFINET interface [X1]\Advanced options\Interface options			
Support device replacement without exchangeable medium	True	Permit overwriting of device names of all assigned IO devices	False
Use IEC V2.2 LLDP mode	False	Keep-Alive connection monitoring:	30s
PROFINET interface [X1]\Advanced options\Real time settings\IO communication			
Send clock:	1.000ms		
PROFINET interface [X1]\Advanced options\Real time settings\Real time options			
Calculated bandwidth for cyclic IO data:	0.000ms	Calculated bandwidth for cyclic IO data:	0.000%
PROFINET interface [X1]\Advanced options\Port [X1 P1]\General			
Name	Port_1	Author	ENVY
Comment			
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Local port:			
Local port:	PLC_Sorting\PROFINET interface_1 [X1]\Port_1 [X1 P1]	Medium:	Copper
Cable name:	---		
			
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port interconnection\Partner port:			
	Monitoring of partner port is not possible	Partner port:	Any partner
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Activate			
Activate this port for use	True		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Connection			
Transmission rate / duplex:	Automatic	Monitor	False
Enable autonegotiation	True		
PROFINET interface [X1]\Advanced options\Port [X1 P1]\Port options\Boundaries			
End of detection of accessible devices	False	End of topology discovery	False
End of the sync domain	False		
PROFINET interface [X1]\Web server access			
Enable Web server for the IP address of this interface	False	The Web server must also be activated in the properties of the PLC.	
High speed counters (HSC)\HSC1\General\Enable			
Enable this high speed counter	0	Enable this high speed counter	0
Enable this high speed counter	0	Enable this high speed counter	0
Enable this high speed counter	0	Enable this high speed counter	0
High speed counters (HSC)\HSC1\General\Project information			
Name	HSC_1	Comment	
Name	HSC_2	Comment	
Name	HSC_3	Comment	

Totally Integrated Automation Portal			
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Name	HSC_4	Comment	
Name	HSC_5	Comment	
Name	HSC_6	Comment	
High speed counters (HSC)\HSC1\I/O addresses\Input addresses			
Start address	1000.0	End address	1003.7
Start address	1004.0	End address	1007.7
Organization block	0	Start address	1008.0
End address	1011.7	Organization block	0
Process image	0	Start address	1012.0
End address	1015.7	Organization block	0
Process image	0	Start address	1016.0
End address	1019.7	Organization block	0
Process image	0	Start address	1020.0
End address	1023.7	Organization block	0
Process image	0	Organization block	0
Process image	0	Process image	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Enable			
Enable this pulse generator	0	Enable this pulse generator	0
Pulse generators (PTO/PWM)\PTO1/PWM1\General\Project information			
Name	Pulse_1	Comment	
Name	Pulse_2	Comment	
Pulse generators (PTO/PWM)\PTO1/PWM1\I/O addresses\Output addresses			
Start address	1000.0	End address	1001.7
Start address	1002.0	End address	1003.7
Organization block	0	Organization block	0
Process image	0	Process image	0
Startup			
Startup after POWER ON	Warm restart - mode before POWER OFF	Comparison preset to actual configuration	Startup CPU even if mismatch
Configuration time	60000ms	OBs should be interruptible	1
Cycle			
Cycle monitoring time [ms]	150ms		
Enable minimum cycle time for cyclic OBs	0	Minimum cycle time	1ms
Communication load			
Cycle load due to communication [%]	20%		
System and clock memory\System memory bits			
Enable the use of system memory byte	0	Address of system memory byte (MBx)	1
First cycle		Diagnostic status changed	
Always 1 (high)		Always 0 (low)	
System and clock memory\Clock memory bits			
Enable the use of clock memory byte	0	Address of clock memory byte (MBx)	0
10 Hz clock		5 Hz clock	
2.5 Hz clock		2 Hz clock	
1.25 Hz clock		1 Hz clock	
0.625 Hz clock		0.5 Hz clock	
Web server\General			
Activate Web server on all modules of this device	False	Permit access only with HTTPS	True

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Totally Integrated Automation Portal					
Web server\Automatic update					
Enable automatic update	True		Update interval	0s	
Web server\User management					
User name			User rights		
Everybody					
Web server\User-defined web pages					
Application name	HTML source path	Default HTML page	Files with dynamic content	Web DB number	Fragment DB number
		index.htm	.htm;.html	333	334
Web server\Overview of interfaces					
Device		Interface		Enabled web server access	
PLC_Sorting		PROFINET interface_1		False	
User interface languages					
Assign project language			User interface languages		
English (United States)			German		
English (United States)			English		
English (United States)			French		
English (United States)			Spanish		
English (United States)			Italian		
English (United States)			Chinese (simplified)		
Time of day\Local time					
Time zone	(UTC +01:00) Berlin, Bern, Brussels, Rome, Stockholm, Vienna				
Time of day\Daylight saving time					
Activate daylight saving time	1		Difference between standard and daylight saving time	60min	
Time of day\Daylight saving time\Start of daylight saving time					
Starting week of the month:	Last			Sunday	
in	March		at	1:00 a.m.	
Time of day\Daylight saving time\Start of standard time					
	Last			Sunday	
in	October		at	2:00 a.m.	
Protection & Security					
Level of protection	No protection				
Protection & Security\Connection mechanisms					
Permit access with PUT/GET communication from remote partner	True				
Protection & Security\Security event					
Summarize diagnostics in case of high message volume	True		Length of an interval	20	
Unit	seconds				
Protection & Security\External load memory					
Disable copying from internal load memory to external load memory	False				
Advanced configuration\DnsParameterConfigurationMenu					
The Tree-Node DnsConfigurationMenu was not filled by some ACF					

Totally Integrated Automation Portal					
Advanced configuration\Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Advanced configuration\SNMP\SNMP configuration (Simple Network Management Protocol).					
Activate SNMP	False				
Configuration control\Configuration control for central configuration					
Allow to reconfigure the device via the user program	0				
Connection resources\					
	Station resources - Reserved - Maximum	Station resources - Reserved - Configured	Station resources - Dynamic - Configured	Module resources - PLC_Sorting [CPU 1214C DC/DC/DC] - Configured	
Maximum number of resources:		34	34	68	
	Maximum	Configured	Configured	Configured	
PG communication:	4	-	-	-	
HMI communication:	12	0	0	0	
S7 communication:	8	0	0	0	
Open user communication:	8	0	0	0	
Web communication:	2	-	-	-	
OPC UA client/server communication:	0	-	-	-	
Other communication:	-	-	0	0	
Total resources used:		0	0	0	
Available resources:		34	34	68	
Overview of addresses\Overview of addresses\Overview of addresses					
Inputs	True		Outputs	True	
Address gaps	False		Slot	True	

Totally Integrated Automation Portal										
Type	Addr. from	Addr. to	Module	PIP	Device name	Device number	Size	Master / IO system	Rack	Slot
I	64	67	AI 2_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 2
I	1000	1003	HSC_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 16
I	1004	1007	HSC_2	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 17
I	1008	1011	HSC_3	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 18
I	1012	1015	HSC_4	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 19
I	1016	1019	HSC_5	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 20
I	1020	1023	HSC_6	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	4 Bytes	-	0	1 21
O	1000	1001	Pulse_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 32
O	1002	1003	Pulse_2	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 33
O	1004	1005	Pulse_3	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 34
O	1006	1007	Pulse_4	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 35
I	100	101	DI 14/DQ 10_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 1
O	0	1	DI 14/DQ 10_1	Automatic update	PLC_Sorting [CPU 1214C DC/DC/DC]	-	2 Bytes	-	0	1 1

Totally Integrated Automation Portal		
SortingLine_Project Ungrouped devices This folder is empty.		

Totally Integrated Automation Portal		
SortingLine_Project Security settings This folder is empty.		

Totally Integrated Automation Portal		
SortingLine_Project / Cross-device functions / Project traces Measurements This folder is empty.		

SortingLine_Project / Common data / Alarm classes

Alarm classes

Alarm classes				
Name	ID	Display name	Acknowledgment	Priority
Acknowledgement	33	A	True	0
No Acknowledgement	34	NA	False	0

Totally Integrated Automation Portal		
SortingLine_Project / Common data Logs This folder is empty.		

Totally Integrated Automation Portal		
SortingLine_Project / Languages & resources Project languages Languages Reference language English (United States) Editing language English (United States) Other project languages Empty		

SortingLine_Project / Languages & resources / Project texts

Project texts

Project texts	
English (United States)	exit left active
Category	Block comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 1\Title
English (United States)	A
Category	Alarm class text
Reference	SortingLine_Project\Acknowledgement\AlarmClassData_IDisplayNaming_DisplayName
English (United States)	A
Category	Alarm class text
Reference	SortingLine_Project\Acknowledgement\ShortName
English (United States)	Abort when not running
Category	Block comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 18\Title
English (United States)	Active (TRUE) for both low and high boxes.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_LowBox\Comment
English (United States)	Active (TRUE) only alongside LowBox for high boxes.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_HighBox\Comment
English (United States)	Active on safety Fault or when system is in Manual mode.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_YellowIndicator\Comment
English (United States)	Active when cycle is offline.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_StopLight\Comment
English (United States)	Active when safety is tripped and waiting for Reset.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_ResetLight\Comment
English (United States)	Auto mode
Category	Block comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 1\Title
English (United States)	Center reflective sensor inside the turntable. Used for precise stopping.
Category	Text category tag comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atTurntableEntry\Comment
English (United States)	Center timer
Category	Block comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 10\Title
English (United States)	Centralized Output Gating and Hardware Interlocks
Category	Block comment
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Block title
English (United States)	chosen entry sensor
Category	Block comment

Totally Integrated Automation Portal		
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 7\Title	
English (United States)	Clear left counter	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 38\Title	
English (United States)	Clear right counter	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 39\Title	
English (United States)	Clears safety fault latches.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_Reset\Comment	
English (United States)	Confirms a box has fully cleared the entry area.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atBack\Comment	
English (United States)	Confirms the box has successfully transitioned onto the Left exit conveyor.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atLeftEntry\Comment	
English (United States)	Confirms the box has successfully transitioned onto the Right exit conveyor.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atRightEntry\Comment	
English (United States)	Constant green indicates system is running active cycles.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_GreenIndicator\Comment	
English (United States)	Constant red indicates system is idle/stopped.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_RedIndicator\Comment	
English (United States)	Continuous cycle	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 25\Title	
English (United States)	Continuous TRUE spawns boxes sequentially.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_Emit\Comment	
English (United States)	Count left	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 36\Title	
English (United States)	Count right	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 37\Title	
English (United States)	Counter display	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 18\Title	
English (United States)	Cycle idle	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 34\Title	
English (United States)	Detects box emergence.	
Category	Text category tag comment	

Totally Integrated Automation Portal		
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atEntry\Comment	
English (United States)	Digital totalizer display unit located on the operator panel.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_Counter\Comment	
English (United States)	Drives the first entry section right beneath the box emitter.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_FeederConveyor\Comment	
English (United States)	Drives the left clearance exit lane.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_LeftConveyor\Comment	
English (United States)	Drives the right clearance exit lane.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_RightConveyor\Comment	
English (United States)	Drives the second section of the entry lane towards the turntable.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_EntryConveyor\Comment	
English (United States)	Drops to FALSE when a box reaches the left remover.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atLeftExit\Comment	
English (United States)	Drops to FALSE when a box reaches the right remover.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atRightExit\Comment	
English (United States)	Edge memory bits	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\s_EdgeStart	
English (United States)	Edge memory bits	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\s_EdgeReset	
English (United States)	Emergency stop	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_EstopNC	
English (United States)	Emit command	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 28\Title	
English (United States)	Emit timer	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 27\Title	
English (United States)	Emitter config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 11\Title	
English (United States)	Enable	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 1\Title	
English (United States)	entry conveyor config	
Category	Block comment	

Totally Integrated Automation Portal		
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 3\Title	
English (United States)	Entry seen latch	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 9\Title	
English (United States)	Exit conveyors	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 33\Title	
English (United States)	exit left pulse	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 3\Title	
English (United States)	exit right active	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 2\Title	
English (United States)	exit right pulse	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 4\Title	
English (United States)	Fault latch	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 5\Title	
English (United States)	Feeder and entry conveyors	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 29\Title	
English (United States)	feeder conveyor config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 2\Title	
English (United States)	Follows the Run state.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_StartLight\Comment	
English (United States)	Global Data Block (DB) serving as the main internal memory exchange registry for the system. Contains structured variables for system operating flags (Run, Fault, Modes), active part counters (Total, Left, Right), manual control parameters generated by the operator panel, and live engineering settings such as the box centering delay time variable.	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\DB_Machine [DB1]\Data block comment	
English (United States)	Global Machine Status and Shared Data Registers	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\DB_Machine [DB1]\Data block title	
English (United States)	Green indicator	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 12\Title	
English (United States)	height latch	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 8\Title	

Totally Integrated Automation Portal		
English (United States)	Implements the continuous automatic sorting sequence using a customized sequential state machine (Steps 0 to 5). Manages entry box generation, calculates sensor filtration via an independent centering timer block, tracks multiple parts in transit using dedicated software registers (In-Transit counters), and automatically routes boxes based on height measurements recorded by the light curtain.	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Block comment	
English (United States)	Instantly deletes boxes at the left exit limit.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_RemoverLeft\Comment	
English (United States)	Instantly deletes boxes at the right exit limit.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_RemoverRight\Comment	
English (United States)	Left conveyor config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 7\Title	
English (United States)	Left entry pulse	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 5\Title	
English (United States)	Left in transit clear	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 21\Title	
English (United States)	Left in transit minus	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 20\Title	
English (United States)	Left in transit plus	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 19\Title	
English (United States)	Load command	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 30\Title	
English (United States)	Load config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 4\Title	
English (United States)	Machine Modes and Safety Interlocking Manager	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Block title	
English (United States)	Main cyclic block of the S7-1200 PLC. This block coordinates the execution of the entire digital twin program at every PLC scan cycle. It handles the mode manager block to evaluate safety states, runs the main automatic sorting logic sequence, and executes the centralized output gating block to control the physical actuators of the Factory I/O scene.	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\Main [OB1]\Block comment	
English (United States)	Main Program Cycle Execution	
Category	Block comment	

Totally Integrated Automation Portal		
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\Main [OB1]\Block title	
English (United States)	Manual mode	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 2\Title	
English (United States)	Monitors physical safety devices and panel inputs to manage operational states. Implements Normally Closed (NC) healthy monitoring for the Emergency Stop button, manages the manual safety fault latch (SR flip-flop), and handles the cycle stop requests (StopPending logic). It ensures a clean segregation between AutoMode and ManualMode to prevent overlapping operational commands.	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Block comment	
English (United States)	NA	
Category	Alarm class text	
Reference	SortingLine_Project\No Acknowledgement\AlarmClassData_IDisplayNaming_DisplayName	
English (United States)	NA	
Category	Alarm class text	
Reference	SortingLine_Project\No Acknowledgement\ShortName	
English (United States)	No box being processed	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_CycleIdle	
English (United States)	One scan pulses	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\t_StartPulse	
English (United States)	One scan pulses	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\t_ResetPulse	
English (United States)	Pipelined Box Sorting Sequence (State Machine)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Block title	
English (United States)	Prevents part collisions.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atFront\Comment	
English (United States)	Red indicator	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 14\Title	
English (United States)	Remover left config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 9\Title	
English (United States)	Remover right config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 10\Title	
English (United States)	Reset light	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 17\Title	
English (United States)	Reset pulse	
Category	Block comment	

Totally Integrated Automation Portal		
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 4\Title	
English (United States)	Reset pushbutton	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_Reset	
English (United States)	Right conveyor config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 8\Title	
English (United States)	Right in transit clear	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 24\Title	
English (United States)	Right in transit minus	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 23\Title	
English (United States)	Right in transit plus	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 22\Title	
English (United States)	Run latch	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 7\Title	
English (United States)	Runs internal turntable rollers backward to eject a box out to the left lane.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_Unload\Comment	
English (United States)	Runs internal turntable rollers forward to pull a box inside.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_Load\Comment	
English (United States)	Selector in Auto	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_Auto	
English (United States)	Selector in Auto position.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_Auto\Comment	
English (United States)	Selector in Manual	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_Manual	
English (United States)	Selector in Manual position.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_Manual\Comment	
English (United States)	Serves as the single source of truth for all physical digital and analog outputs directed to the Factory I/O simulation panel. It gathers automatic commands generated by the state machine and manual commands from the watch table. It applies strict hardware safety interlocks (System Enable bit derived from the Emergency Stop state) before forcing any physical coil to prevent duplicate assignment conflicts.	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Block comment	
English (United States)	Start light	

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Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 13\Title	
English (United States)	Start pulse	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 3\Title	
English (United States)	Start pushbutton	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_Start	
English (United States)	Step output	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 35\Title	
English (United States)	Stop at end of cycle	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 26\Title	
English (United States)	Stop light	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 15\Title	
English (United States)	Stop pending	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\Network 6\Title	
English (United States)	Stop pushbutton	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_ModeManager [FB1]\i_StopNC	
English (United States)	Swivels the turntable mechanism 90° clockwise.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Outputs [17]\Q_Turn\Comment	
English (United States)	T0 to 1 (#s_Step := 1)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 15\Title	
English (United States)	T1 to 2 (#s_Step := 2)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 14\Title	
English (United States)	T2 to 3 (#s_Step := 3)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 13\Title	
English (United States)	T3 to 4 (#s_Step := 4)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 12\Title	
English (United States)	T4 to 5 (#s_Step := 5)	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 11\Title	
English (United States)	Total count	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 40\Title	
English (United States)	Triggers the cycle run latch.	

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Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_Start\Comment	
English (United States)	TRUE at rest, drops to FALSE to request a cycle stop.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_Stop\Comment	
English (United States)	TRUE when healthy, drops to FALSE when pressed.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_EmergencyStop\Comment	
English (United States)	TRUE when turntable has completed its 90° clockwise rotation.	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atUnloadPosition\Comment	
English (United States)	TRUE when turntable is at 0° (aligned with entry).	
Category	Text category tag comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\PLC tags\Factory_IO_Inputs [18]\I_atLoadPosition\Comment	
English (United States)	Turnr config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 6\Title	
English (United States)	Turn command	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 32\Title	
English (United States)	Unload command	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FB_SortingLogic [FB2]\Network 31\Title	
English (United States)	Unload config	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 5\Title	
English (United States)	Yellow indicator	
Category	Block comment	
Reference	SortingLine_Project\PLC_Sorting [CPU 1214C DC/DC/DC]\Program blocks\FC_outputs [FC1]\Network 16\Title	